

CAMBRIDGE INTERNATIONAL AS &amp; A LEVEL CHEMISTRY (9701)

# Topic 19 — Nitrogen Compounds: Primary Amines, Amides & Nitriles (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions &amp; Detailed Solutions

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|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| CANDIDATE      | Urwah                                                                                                                           |
| SYLLABUS       | Cambridge International AS Level Chemistry (9701)                                                                               |
| COMPONENT      | Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)                                                                      |
| TOPIC COVERAGE | 19.1 Primary Aliphatic Amines: Basicity, Synthesis & Reactions · 19.2 Nitriles & Amides: Formation, Hydrolysis & Neutralisation |
| TOTAL VOLUME   | 110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)                                                               |

## HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

## SYLLABUS SUBTOPICS COVERED IN THIS PACK

- 19.1 Primary Aliphatic Amines: Basicity, Synthesis & Reactions** — Basicity and pyramidal geometry ( $\sim 107^\circ$ ); intermolecular hydrogen bonding and volatility trends; Bronsted-Lowry acid-base reactions; synthesis from nitriles and aldehydes.
- 19.2 Nitriles & Amides: Formation, Hydrolysis & Neutralisation** — Nitrile group structure, coplanarity, and neutrality in water due to carbonyl pi resonance delocalization; preparation and hydrolysis of nitriles and amides.
- High-Frequency Core Repeats** — The 10 most frequently examined Cambridge Paper 1 questions on Nitrogen Compounds (Q101 – Q110).

## SUBTOPIC 19.1 — PRIMARY ALIPHATIC AMINES: BASICITY, SYNTHESIS &amp; REACTIONS (Q1 – Q55)

## 1 [HARD] Relative Basicity of Nitrogen Compounds — 9701/11/M/J/23/Q32

[1]

Which sequence correctly ranks the following nitrogen-containing compounds in order of DECREASING base strength (strongest base to weakest base)?

- [A] Phenylamine > Ammonia > Ethylamine
- [B] Ethylamine ( $\text{CH}_3\text{CH}_2\text{NH}_2$ ) > Ammonia ( $\text{NH}_3$ ) > Phenylamine ( $\text{C}_6\text{H}_5\text{NH}_2$ )
- [C] Ammonia > Ethylamine > Phenylamine
- [D] Ethylamine > Phenylamine > Ammonia

## 2 [EASY] Intermolecular Hydrogen Bonding in Amines vs Alcohols — 9701/12/M/J/23/Q32

[1]

Ethylamine ( $M_r = 45.0$ ) boils at  $16.6^\circ\text{C}$ , whereas ethanol ( $M_r = 46.0$ ) boils at  $78.4^\circ\text{C}$ .

What explains why the boiling point of ethylamine is so much lower than that of ethanol?

- [A] Ethylamine cannot form intermolecular hydrogen bonds.
- [B] Ethanol forms covalent bonds with nitrogen in the air.
- [C] Ethylamine has a smaller electron cloud and weaker London dispersion forces.
- [D] Nitrogen is less electronegative than oxygen (3.0 vs 3.5), making the N-H bond less polar than the O-H bond, resulting in significantly weaker intermolecular hydrogen bonds in amines than in alcohols.

## 3 [MEDIUM] Synthesis of Pure Primary Amines via Nitrile Reduction — 9701/13/M/J/23/Q32

[1]

Why is the reduction of a nitrile (e.g. propanenitrile with  $\text{LiAlH}_4$ ) preferred over reacting 1-bromopropane with ammonia for preparing propan-1-amine?

- [A] Nitrile reduction yields pure primary amine with zero contamination by secondary or tertiary amines or quaternary salts.
- [B] Nitrile reduction does not require any solvent.
- [C] 1-bromopropane does not react with ammonia under any conditions.
- [D] Nitriles are completely non-toxic and natural.

## 4 [EASY] Reaction of Amines with Mineral Acids — 9701/11/O/N/23/Q32

[1]

What is observed when ethylamine vapor is brought into contact with hydrogen chloride gas, and what is the nature of the product?

- [A] A brown gas of nitrogen dioxide is evolved.
- [B] A colorless liquid ester is formed.
- [C] Dense white fumes of the ionic salt ethylammonium chloride,  $\text{CH}_3\text{CH}_2\text{NH}_3^+ \text{Cl}^-$ .
- [D] Effervescence of hydrogen gas.

## 5 [EASY] Molecular Geometry Around Amine Nitrogen — 9701/13/M/J/22/Q5

[1]

What is the molecular geometry and approximate bond angle around the nitrogen atom in methylamine ( $\text{CH}_3\text{NH}_2$ )?

- [A] Trigonal pyramidal with bond angle approximately  $107^\circ$
- [B] Trigonal planar with bond angle  $120^\circ$
- [C] Tetrahedral with bond angle  $109.5^\circ$
- [D] Linear with bond angle  $180^\circ$

## 6 [HARD] Water Solubility of Primary Aliphatic Amines — 9701/11/M/J/22/Q6

[1]

Why are methylamine, ethylamine, and propylamine completely miscible with water, producing alkaline solutions?

- [A] They react with water to form explosive hydrogen gas.
- [B] They are non-polar liquids.
- [C] They precipitate as insoluble hydroxides.
- [D] They form extensive intermolecular hydrogen bonds with water molecules and partially ionize to liberate hydroxide ions:  $\text{RNH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{RNH}_3^+ + \text{OH}^-$ .

**7 [EASY]** Suppression of Polyalkylation with Ammonia — 9701/12/M/J/222/Q7**[1]**

In the preparation of ethylamine from bromoethane and ammonia, why must a LARGE EXCESS of concentrated ammonia be employed?

- [A] To increase the temperature of the reaction.
- [B] To maximize the probability of bromoethane colliding with ammonia molecules rather than with newly formed ethylamine molecules, suppressing secondary and tertiary amine formation.
- [C] To neutralize the sulfuric acid catalyst.
- [D] To act as an inert solvent.

**8 [HARD]** Sealed Tube Reaction for Volatile Amines — 9701/13/M/J/223/Q8**[1]**

Why must the reaction between bromoethane and ammonia be carried out in a sealed tube under pressure?

- [A] To prevent oxygen from igniting the bromoethane.
- [B] To compress the atoms into a polymer.
- [C] Ammonia and bromoethane are highly volatile gases/liquids that would escape from an open reflux condenser at the reaction temperature (100 °C).
- [D] To create a vacuum inside the flask.

**9 [EASY]** Basicity of Secondary Amines vs Primary Amines — 9701/11/M/J/224/Q9**[1]**

Why are secondary aliphatic amines (such as dimethylamine,  $(\text{CH}_3)_2\text{NH}$ ) slightly stronger bases in aqueous solution than primary amines (methylamine,  $\text{CH}_3\text{NH}_2$ )?

- [A] Secondary amines have two lone pairs on nitrogen.
- [B] Two electron-donating (+I) alkyl groups push more electron density onto nitrogen than one alkyl group, increasing the availability of the lone pair.
- [C] Secondary amines form fewer hydrogen bonds with water.
- [D] Secondary amines are completely ionic.

**10 [HARD]** Basicity of Tertiary Amines in Aqueous Solution — 9701/12/M/J/220/Q10**[1]**

Why is triethylamine,  $(\text{C}_2\text{H}_5)_3\text{N}$ , slightly WEAKER as a base in aqueous solution than diethylamine,  $(\text{C}_2\text{H}_5)_2\text{NH}$ , despite having three +I ethyl groups?

- [A] Steric hindrance from three bulky ethyl groups impedes the approach of a proton and reduces stabilizing hydration of the protonated cation by water molecules.
- [B] Tertiary amines have no non-bonding lone pairs.
- [C] The C-N bonds in tertiary amines are pure ionic.
- [D] Triethylamine decomposes in water.

**11 [EASY]** Basicity of Phenylamine (Aniline) — 9701/13/M/J/221/Q11**[1]**

Why is phenylamine ( $\text{C}_6\text{H}_5\text{NH}_2$ ,  $\text{pK}_\text{b} = 9.4$ ) an extraordinarily weak base compared to ammonia ( $\text{pK}_\text{b} = 4.75$ )?

- [A] Phenylamine is an aliphatic amine.
- [B] The phenyl ring repels protons by magnetic forces.
- [C] Phenylamine dissociates into cyanide ions.
- [D] The unshared electron pair on the nitrogen atom delocalises into the aromatic pi system of the benzene ring, making it unavailable to accept a proton.

**12 [HARD]** Protonation of Phenylamine: Loss of Resonance — 9701/11/M/J/222/Q12**[1]**

Why is the protonation of phenylamine ( $\text{C}_6\text{H}_5\text{NH}_2 + \text{H}^+ \rightarrow \text{C}_6\text{H}_5\text{NH}_3^+$ ) thermodynamically unfavorable?

- [A] The benzene ring breaks apart upon protonation.
- [B] Anilinium ions are explosive.
- [C] Protonation binds the nitrogen lone pair into an N-H sigma bond, completely destroying its resonance delocalization into the benzene ring and losing  $\sim 30 \text{ kJ mol}^{-1}$  of resonance stabilization energy.
- [D] Phenylamine cannot dissolve in acid.

**13** [EASY] Reaction of Amines with Copper(II) Ions — 9701/12/M/J/223/Q13

[1]

What is observed when a small amount of aqueous ethylamine is added to aqueous copper(II) sulfate, followed by excess ethylamine?

- [A] Initially a pale blue precipitate of  $\text{Cu}(\text{OH})_2$  forms, which dissolves in excess ethylamine to form a deep blue solution of a copper-amine complex.
- [B] A shiny mirror of copper metal.
- [C] A yellow precipitate of sulfur.
- [D] A bright green solution of copper chloride.

**14** [HARD] Amines as Nucleophiles: Reaction with Haloalkanes — 9701/13/M/J/224/Q14

[1]

What type of reaction occurs between ethylamine and bromoethane to form diethylamine?

- [A] Electrophilic addition.
- [B] Free radical substitution.
- [C] Nucleophilic substitution (the lone pair on nitrogen attacks the  $\delta^+$  carbon of bromoethane).
- [D] Elimination.

**15** [EASY] Quaternary Ammonium Salts: Structure and Uses — 9701/11/M/J/220/Q15

[1]

What is the structure of tetramethylammonium bromide, and what is its electrical conductivity in water?

- [A] A neutral covalent molecule with no electrical conductivity.
- [B] An ionic salt consisting of  $(\text{CH}_3)_4\text{N}^+$  cations and  $\text{Br}^-$  anions; completely dissociated and conducts electricity strongly in aqueous solution.
- [C] A polymer of methylamine.
- [D] A coordination compound of iron.

**16** [HARD] Cationic Surfactants from Quaternary Ammonium Salts — 9701/12/M/J/221/Q16

[1]

Why are long-chain quaternary ammonium salts (such as cetyltrimethylammonium bromide) used in hair conditioners and fabric softeners?

- [A] They bleach stains using chlorine.
- [B] They dissolve hair fibers completely.
- [C] They act as strong acids that dissolve grease.
- [D] The positively charged quaternary head binds electrostatically to negatively charged protein/fabric fibers, while the oily hydrocarbon tail smooths the surface.

**17** [EASY] Distinguishing Primary, Secondary, and Tertiary Amines — 9701/13/M/J/222/Q17

[1]

How do primary, secondary, and tertiary aliphatic amines behave when treated with ethanoyl chloride ( $\text{CH}_3\text{COCl}$ )?

- [A] All three form identical amides.
- [B] Only tertiary amines react.
- [C] Primary and secondary amines react vigorously to form N-substituted amides and steamy  $\text{HCl}$  fumes; tertiary amines cannot form amides because they lack an  $\text{N-H}$  bond.
- [D] None of them react with acyl chlorides.

**18** [HARD] Odor of Amines — 9701/11/M/J/223/Q18

[1]

What characteristic odor is associated with volatile aliphatic amines (such as trimethylamine and 1,5-diaminopentane)?

- [A] A sharp, pungent, unpleasant fishy or decaying flesh odor.
- [B] A sweet, fruity perfume odor.
- [C] A suffocating choking sulfur smell.
- [D] Completely odorless.

**19** [EASY] Extraction of Amines using Acid-Base Chemistry — 9701/12/M/J/224/Q19**[1]**

How can an organic base like ethylamine be separated from a mixture with neutral hexane in a separating funnel?

- [A] Shake with water only.
- [B] Boil the mixture at 200 °C.
- [C] Add sodium metal.
- [D] Shake with dilute hydrochloric acid: ethylamine reacts to form water-soluble ethylammonium chloride in the aqueous layer, which is separated and treated with NaOH to regenerate pure ethylamine.

**20** [HARD] Combustion of Amines — 9701/13/M/J/220/Q20**[1]**

What gases are produced by the complete combustion of methylamine,  $\text{CH}_3\text{NH}_2$ , in excess oxygen?

- [A] Carbon monoxide and ammonia
- [B] Carbon dioxide ( $\text{CO}_2$ ), water vapor ( $\text{H}_2\text{O}$ ), and nitrogen gas ( $\text{N}_2$ )
- [C] Nitric acid and methane
- [D] Hydrogen cyanide and oxygen

**21** [EASY] Chirality in Amines: Pyramidal Inversion — 9701/11/M/J/221/Q21**[1]**

Why can an unsymmetrical tertiary amine with three different alkyl groups (e.g. N-ethyl-N-methylpropan-1-amine) NOT be resolved into stable optical enantiomers at room temperature?

- [A] The C-N bonds break spontaneously.
- [B] The nitrogen atom undergoes rapid umbrella-like pyramidal inversion (inversion of the lone pair) at room temperature with a very low activation energy barrier ( $\sim 25 \text{ kJ mol}^{-1}$ ).
- [C] Amines are completely planar.
- [D] Amines only exist as racemic crystals.

**22** [HARD] Preparation of Aromatic Amines via Nitrobenzene Reduction — 9701/12/M/J/222/Q22**[1]**

What reagents and conditions are required to reduce nitrobenzene ( $\text{C}_6\text{H}_5\text{NO}_2$ ) to phenylamine ( $\text{C}_6\text{H}_5\text{NH}_2$ )?

- [A] Sodium borohydride in water.
- [B] Hydrogen gas at room temperature without catalyst.
- [C] Acidified potassium dichromate.
- [D] Heating under reflux with tin (Sn) and concentrated hydrochloric acid, followed by addition of excess aqueous sodium hydroxide.

**23** [EASY] Bromination of Phenylamine — 9701/13/M/J/223/Q23**[1]**

What is observed when bromine water is added dropwise to an aqueous solution of phenylamine at room temperature?

- [A] The orange-brown bromine water is decolorized, and a white precipitate of 2,4,6-tribromophenylamine is formed immediately without requiring a catalyst.
- [B] A purple vapor is evolved.
- [C] A silver mirror forms.
- [D] No reaction occurs.

**24** [HARD] Diazotisation of Phenylamine — 9701/11/M/J/224/Q24**[1]**

What reagents and temperature are required to convert phenylamine into benzenediazonium chloride, and why must the temperature be kept below 5 °C?

- [A] Nitric acid at 100 °C.
- [B] Ammonia and hydrogen peroxide at 50 °C.
- [C] Sodium nitrite ( $\text{NaNO}_2$ ) and concentrated HCl between 0 °C and 5 °C; temperatures above 5 °C decompose the diazonium salt into phenol and nitrogen gas.
- [D] Chlorine gas in UV light at 25 °C.

**25** [EASY] Azo Dye Formation (Coupling Reaction) — 9701/12/M/J/220/Q25

[1]

What type of colored compound is formed when cold benzenediazonium chloride is reacted with an alkaline solution of phenol?

- [A] A brightly colored yellow/orange azo dye (4-hydroxyphenylazobenzene) containing the -N=N- azo chromophore.
- [B] A colorless polymer.
- [C] A dense black precipitate of carbon.
- [D] A sweet-smelling ester.

**26** [HARD] Primary Aliphatic vs Aromatic Diazonium Ions — 9701/13/M/J/221/Q26

[1]

Why are aliphatic diazonium ions (e.g.  $\text{CH}_3\text{CH}_2\text{N}_2^+$ ) completely unstable even at  $0^\circ\text{C}$ , while benzenediazonium ions ( $\text{C}_6\text{H}_5\text{N}_2^+$ ) are stable below  $5^\circ\text{C}$ ?

- [A] Aliphatic diazonium ions are solid polymers.
- [B] Benzenediazonium ions contain no nitrogen atoms.
- [C] Aliphatic diazonium ions have higher molecular masses.
- [D] In benzenediazonium ions, the positive charge and diazonium pi electrons are stabilized by resonance delocalisation into the aromatic benzene ring.

**27** [EASY] Basicity Comparison: Cyclohexylamine vs Phenylamine — 9701/11/M/J/222/Q27

[1]

Why is cyclohexylamine ( $\text{pK}_a$  of conjugate acid = 10.6) a million times stronger base than phenylamine ( $\text{pK}_a$  of conjugate acid = 4.6)?

- [A] Cyclohexylamine has fewer carbon atoms.
- [B] Cyclohexylamine is a non-aromatic, saturated aliphatic amine with an  $\text{sp}^3$  ring that does not delocalise the nitrogen lone pair, whereas in phenylamine the lone pair is delocalised into the aromatic ring.
- [C] Phenylamine is a mineral acid.
- [D] Cyclohexylamine has no nitrogen lone pair.

**28** [HARD] Infrared Spectrum of Primary Amines — 9701/12/M/J/223/Q28

[1]

What characteristic absorption bands identify a primary aliphatic amine in an infrared spectrum?

- [A] A single sharp peak at  $1710\text{ cm}^{-1}$ .
- [B] A broad band from  $2500$  to  $3300\text{ cm}^{-1}$ .
- [C] A doublet (two distinct absorption spikes) in the N-H stretching region between  $3300$  and  $3500\text{ cm}^{-1}$ .
- [D] No peaks above  $2000\text{ cm}^{-1}$ .

**29** [EASY] Boiling Point: Primary vs Secondary vs Tertiary Amines — 9701/13/M/J/224/Q29

[1]

Among isomeric amines of formula  $\text{C}_3\text{H}_9\text{N}$  (propylamine, ethylmethylamine, and trimethylamine), which has the LOWEST boiling point and why?

- [A] Propylamine (bp  $48^\circ\text{C}$ ), because it is linear.
- [B] Trimethylamine (bp  $3^\circ\text{C}$ ), because it has no N-H bond and cannot form intermolecular hydrogen bonds with itself.
- [C] Ethylmethylamine (bp  $36^\circ\text{C}$ ), because it is secondary.
- [D] All three have identical boiling points.

**30** [HARD] Reaction of Amines with Carbon Dioxide — 9701/11/M/J/220/Q30

[1]

Why does an aqueous solution of ethylamine slowly absorb carbon dioxide from the air to form a white crystalline crust?

- [A] Ethylamine is basic and reacts with weakly acidic  $\text{CO}_2$  to form ethylammonium ethylcarbamate salt:  $2\text{C}_2\text{H}_5\text{NH}_2 + \text{CO}_2 \rightarrow \text{C}_2\text{H}_5\text{NH}_3^+ [\text{C}_2\text{H}_5\text{NHCOO}]^-$ .
- [B] Ethylamine burns in  $\text{CO}_2$ .
- [C] Carbon dioxide reduces ethylamine to ethane.
- [D] Ethylamine polymerises into nylon.

**31** [EASY] Synthesis of Urea — 9701/12/M/J/221/Q31**[1]**

What is urea, and how is it synthesized industrially from ammonia and carbon dioxide?

- [A] Urea is a polymer of amino acids.
- [B] Urea is an ester of nitric acid.
- [C] Urea is made by cracking petroleum.
- [D] Urea is the diamide of carbonic acid,  $\text{H}_2\text{N}-\text{CO}-\text{NH}_2$ ; synthesized by reacting liquid ammonia with carbon dioxide at  $200^\circ\text{C}$  and 200 atm pressure.

**32** [HARD] Solubility of Amine Hydrochlorides — 9701/13/M/J/222/Q32**[1]**

Why are high-molecular-weight pharmaceutical drugs containing amine groups (such as morphine or pseudoephedrine) formulated as hydrochloride salts?

- [A] Hydrochloride salts are non-toxic gases.
- [B] Amines explode unless salted.
- [C] Converting the neutral insoluble amine into an ionic alkylammonium chloride salt vastly increases water solubility and bioavailability in the bloodstream.
- [D] Chlorides improve the taste.

**33** [EASY] Reaction of Amines with Nitrous Acid (Aliphatic) — 9701/11/M/J/223/Q33**[1]**

What is observed when a primary aliphatic amine (e.g. ethylamine) reacts with cold nitrous acid ( $\text{NaNO}_2 + \text{dilute HCl}$  at  $0^\circ\text{C}$ )?

- [A] Vigorous effervescence of nitrogen gas ( $\text{N}_2$ ) and formation of an alcohol (ethanol).
- [B] A permanent yellow diazonium solution.
- [C] A white precipitate of tribromoethylamine.
- [D] No reaction occurs.

**34** [HARD] Distinguishing Aliphatic from Aromatic Amines with  $\text{HNO}_2$  — 9701/12/M/J/224/Q34**[1]**

How does the reaction with cold nitrous acid ( $0-5^\circ\text{C}$ ) distinguish ethylamine from phenylamine?

- [A] Phenylamine evolves oxygen gas.
- [B] Ethylamine turns blue.
- [C] Ethylamine immediately evolves nitrogen gas with vigorous bubbling; phenylamine forms a stable benzenediazonium salt that does not evolve gas and forms an orange azo dye with phenol.
- [D] Both evolve hydrogen gas.

**35** [EASY] Basicity in Gas Phase vs Aqueous Solution — 9701/13/M/J/220/Q35**[1]**

In the gas phase (absence of solvent), what is the true intrinsic order of base strength for methylamines?

- [A] Ammonia > Methylamine > Dimethylamine > Trimethylamine
- [B] Trimethylamine > Dimethylamine > Methylamine > Ammonia
- [C] Methylamine > Dimethylamine > Trimethylamine
- [D] All four have identical intrinsic basicity.

**36** [HARD] Alkylammonium Ion Charge Dispersal — 9701/11/M/J/221/Q36**[1]**

How does the alkylammonium cation ( $\text{R}-\text{NH}_3^+$ ) stabilize its positive charge compared to the ammonium ion ( $\text{NH}_4^+$ )?

- [A] The alkyl group radiates energy.
- [B] The positive charge is neutralized by carbon.
- [C] Alkyl groups form covalent bonds with solvent.
- [D] The alkyl group disperses the positive charge by pushing electron density toward nitrogen (+I effect), lowering the electrostatic potential energy of the ion.

**37** [EASY] Amine Nucleophilicity vs Basicity — 9701/12/M/J/222/Q37

[1]

Why is triethylamine a weaker nucleophile than quinuclidine (a bicyclic tertiary amine with locked back-facing alkyl groups)?

- [A] Quinuclidine is an aromatic compound.
- [B] Triethylamine contains no lone pairs.
- [C] In triethylamine, the three freely rotating ethyl groups sterically hinder access to the nitrogen lone pair, whereas in quinuclidine the alkyl cage is locked backwards, fully exposing the lone pair.
- [D] Triethylamine has lower molecular weight.

**38** [HARD] Synthesis of Amines via Azide Reduction — 9701/13/M/J/223/Q38

[1]

What sequence converts 1-chlorobutane into butan-1-amine without polyalkylation?

- [A] React with sodium azide ( $\text{NaN}_3$ ) in ethanol to form 1-azidobutane ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{N}_3$ ), followed by reduction with  $\text{LiAlH}_4$  or  $\text{H}_2/\text{Pd}$ .
- [B] React with ammonia and chlorine.
- [C] React with nitric acid and tin.
- [D] Heat with ammonium chloride.

**39** [EASY] Gabriel Phthalimide Synthesis Overview — 9701/11/M/J/224/Q39

[1]

What is the primary advantage of the Gabriel phthalimide synthesis for preparing primary aliphatic amines?

- [A] It does not require heating.
- [B] It uses petroleum directly.
- [C] It works only for aromatic amines.
- [D] It selectively produces 100% pure primary amines with zero contamination by secondary or tertiary amines.

**40** [HARD] Reaction of Amines with Carbonyls (Schiff Base Formation) — 9701/12/M/J/220/Q40

[1]

What type of product is formed when a primary amine reacts with an aldehyde or ketone?

- [A] An amide
- [B] An imine (Schiff base),  $\text{R}-\text{CH}=\text{N}-\text{R}'$ , and water (condensation reaction)
- [C] An ester
- [D] A carboxylic acid

**41** [EASY] Reaction of Amines with Chloroform and Alkali (Carbylamine Test) — 9701/13/M/J/221/Q41

[1]

What is the diagnostic carbylamine test for primary amines, and what observation confirms a positive result?

- [A] A silver mirror forms.
- [B] Heating with trichloromethane ( $\text{CHCl}_3$ ) and alcoholic  $\text{KOH}$  produces an intensely foul, nauseating odor of an isocyanide (carbylamine,  $\text{R}-\text{N}\equiv\text{C}$ ).
- [C] A pleasant sweet banana aroma.
- [D] Vigorous effervescence of hydrogen.

**42** [HARD] Hinsberg Test for Amine Classification — 9701/11/M/J/222/Q42

[1]

In the Hinsberg test, an amine is shaken with benzenesulfonyl chloride ( $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ ) and aqueous  $\text{KOH}$ . How does a primary amine behave?

- [A] It forms a solid that is insoluble in  $\text{KOH}$ .
- [B] It produces no reaction.
- [C] It evolves nitrogen gas.
- [D] It forms a sulfonamide that dissolves completely in aqueous  $\text{KOH}$  to form a clear solution (due to acidic  $\text{N}-\text{H}$  proton), which precipitates upon acidification.

**43** [EASY] Hinsberg Test: Secondary vs Tertiary Amines — 9701/12/M/J/223/Q43**[1]**

How do secondary and tertiary amines behave in the Hinsberg test?

- [A] Secondary amines form a neutral solid sulfonamide insoluble in aqueous KOH; tertiary amines show no reaction with benzenesulfonyl chloride.
- [B] Secondary amines dissolve in KOH; tertiary amines form precipitates.
- [C] Both dissolve in acid.
- [D] Neither reacts.

**44** [HARD] Oxidation of Primary Amines — 9701/13/M/J/224/Q44**[1]**

What product is formed when a primary aliphatic amine is oxidized by strong oxidizing agents (such as hydrogen peroxide or  $\text{KMnO}_4$ )?

- [A] Carboxylic acids exclusively.
- [B] Alkenes and nitrogen.
- [C] Oximes or nitroalkanes ( $\text{R-NO}_2$ ) depending on conditions.
- [D] Cyanide gas.

**45** [EASY] Identification of Amines by Elemental Combustion Analysis — 9701/11/M/J/220/Q45**[1]**

A volatile organic base contains 53.3% carbon, 15.6% hydrogen, and 31.1% nitrogen by mass. Its molecular ion peak occurs at  $m/z = 45$ . What is the compound?

- [A] Ethylamine,  $\text{CH}_3\text{CH}_2\text{NH}_2$  (or dimethylamine,  $(\text{CH}_3)_2\text{NH}$ )
- [B] Methylamine
- [C] Propylamine
- [D] Methanamide

**46** [HARD] Water Solubility Trend with Amine Chain Length — 9701/12/M/J/221/Q46**[1]**

Why does the solubility of primary aliphatic amines in water decrease markedly as the carbon chain length increases from methylamine to hexan-1-amine?

- [A] Higher amines decompose in water to form insoluble alkanes.
- [B] Longer carbon chains carry negative charges that repel water.
- [C] Nitrogen becomes positively charged in higher amines.
- [D] The increasingly large hydrophobic hydrocarbon tail disrupts the water hydrogen-bonding network without forming compensating hydrogen bonds.

**47** [EASY] Polyalkylation in Reaction of Haloalkanes with Amines — 9701/13/M/J/222/Q47**[1]**

When ethylamine reacts with bromoethane in a 1:1 molar ratio, why is a mixture of ethylamine, diethylamine, triethylamine, and tetraethylammonium bromide obtained?

- [A] The reaction proceeds via free radical chain branching.
- [B] The ethylamine product remains nucleophilic and competes with unreacted ammonia/amine for bromoethane in consecutive nucleophilic substitution steps.
- [C] Bromoethane eliminates  $\text{HBr}$  which decomposes the amine.
- [D] Ethylamine isomerizes into diethylamine at room temperature.

**48** [HARD] Reaction of Primary Amines with Acid Anhydrides — 9701/11/M/J/223/Q48**[1]**

What products are formed when ethylamine reacts with ethanoic anhydride at room temperature?

- [A] Ethanamide and ethanol
- [B] Diethylamine and water
- [C] N-ethylethanamide ( $\text{CH}_3\text{CONHCH}_2\text{CH}_3$ ) and ethanoic acid ( $\text{CH}_3\text{COOH}$ )
- [D] Ethyl ethanoate and ammonia

49 [EASY] Boiling Point Comparison: Primary vs Secondary Amine Isomers — 9701/12/M/J/224/Q49

[1]

Why does ethylamine ( $\text{CH}_3\text{CH}_2\text{NH}_2$ , bp  $16.6^\circ\text{C}$ ) have a higher boiling point than its constitutional isomer dimethylamine ( $(\text{CH}_3)_2\text{NH}$ , bp  $7.4^\circ\text{C}$ )?

- [A] Dimethylamine cannot form any hydrogen bonds.
- [B] Ethylamine has two hydrogen atoms bonded to nitrogen and can form a more extensive intermolecular hydrogen-bonded network than dimethylamine, which has only one N-H bond.
- [C] Ethylamine has a larger molecular mass.
- [D] The C-N bond in dimethylamine is weaker.

50 [HARD] Basicity Comparison: Amines vs Alcohols and Water — 9701/13/M/J/220/Q50

[1]

Why are aliphatic amines significantly stronger Bronsted-Lowry bases than alcohols or water?

- [A] Nitrogen is less electronegative than oxygen, holding its lone pair of electrons less tightly and making it more readily donated to a proton.
- [B] Amines contain more hydrogen atoms.
- [C] Nitrogen forms quadruple bonds with protons.
- [D] Alcohols are ionic solids.

51 [EASY] Reaction of Diamines with Dihaloalkanes — 9701/11/M/J/221/Q51

[1]

What type of macromolecular or cyclic structure can form when 1,2-diaminoethane reacts with 1,2-dibromoethane in alkaline solution?

- [A] Polyester plastics.
- [B] Addition polymers containing C=C bonds.
- [C] Insoluble metallic crystals.
- [D] Cyclic diamines (such as piperazine derivatives) and linear polyamines via repeated nucleophilic substitutions.

52 [HARD] Reduction of Amides vs Reduction of Nitriles — 9701/12/M/J/222/Q52

[1]

Why does the reduction of ethanamide with  $\text{LiAlH}_4$  yield ethylamine, whereas reduction of ethanenitrile with  $\text{LiAlH}_4$  also yields ethylamine?

- [A] Ethanamide first converts into ethanenitrile in dry ether.
- [B] Both reagents eliminate nitrogen gas.
- [C] Both reactions reduce the functional group carbon atom to a  $-\text{CH}_2-$  methylene unit attached to nitrogen:  $\text{CH}_3\text{CONH}_2 + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{O}$  and  $\text{CH}_3\text{CN} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2$ .
- [D] Both reactions produce ethanol as a byproduct.

53 [EASY] Deprotonation of Alkylammonium Salts — 9701/13/M/J/223/Q53

[1]

What reagent is added to an aqueous solution of ethylammonium chloride,  $\text{CH}_3\text{CH}_2\text{NH}_3^+ \text{Cl}^-$ , to release free ethylamine?

- [A] Aqueous sodium hydroxide ( $\text{NaOH}$ ) or another strong base.
- [B] Dilute hydrochloric acid.
- [C] Acidified potassium dichromate.
- [D] Sodium chloride solution.

54 [HARD] Basicity of Amines in Non-Polar Solvents — 9701/11/M/J/224/Q54

[1]

In a non-polar solvent such as chlorobenzene where hydration effects are negligible, what is the observed order of basicity among methylamines?

- [A] Ammonia > Methylamine > Dimethylamine > Trimethylamine
- [B] Methylamine > Dimethylamine > Trimethylamine
- [C] Trimethylamine > Dimethylamine > Methylamine > Ammonia
- [D] All four have identical basicity in non-polar solvents.

55 [EASY] Amine Coordination with Transition Metal Cations — 9701/12/M/J/220/Q55

[1]

What occurs when a dilute solution of ethylamine is slowly added to an aqueous solution of silver nitrate,  $\text{AgNO}_3$ ?

- [A] A bright green precipitate of silver metal.
- [B] A pale brown precipitate of silver oxide forms initially, which dissolves in excess ethylamine to form the colorless complex diamminesilver-like cation  $[\text{Ag}(\text{CH}_3\text{CH}_2\text{NH}_2)_2]^+$ .
- [C] A yellow precipitate of silver iodide.
- [D] Vigorous fizzing of nitrogen dioxide gas.

**SUBTOPIC 19.2 — NITRILES & AMIDES: FORMATION, HYDROLYSIS & NEUTRALITY (Q56 – Q100)**

56 [HARD] Neutral Nature of Amides in Aqueous Solution — 9701/11/M/J/23/Q33

[1]

Why is ethanamide ( $\text{CH}_3\text{CONH}_2$ ) completely neutral in aqueous solution ( $\text{pH} = 7$ ), whereas ethylamine ( $\text{CH}_3\text{CH}_2\text{NH}_2$ ) is strongly basic ( $\text{pH} = 11-12$ )?

- [A] Amides contain no nitrogen lone pair of electrons.
- [B] Ethanamide dissociates completely into ethanoate and ammonium ions in water.
- [C] The carbonyl oxygen repels all protons by magnetic force.
- [D] The non-bonding lone pair of electrons on the nitrogen atom in ethanamide is delocalised into the adjacent carbonyl  $\pi$  system via resonance, rendering it unavailable to accept a proton.

57 [EASY] Acid-Catalyzed Hydrolysis of Nitriles — 9701/12/M/J/23/Q33

[1]

What are the products formed when propanenitrile ( $\text{CH}_3\text{CH}_2\text{CN}$ ) is heated under reflux with dilute hydrochloric acid?

- [A] Propan-1-amine and chlorine gas.
- [B] Propanal and ammonia.
- [C] Propanoic acid ( $\text{CH}_3\text{CH}_2\text{COOH}$ ) and ammonium chloride ( $\text{NH}_4\text{Cl}$ ).
- [D] Ethyl ethanoate and water.

58 [EASY] Alkaline Hydrolysis of Amides — 9701/13/M/J/23/Q33

[1]

What observation occurs when ethanamide,  $\text{CH}_3\text{CONH}_2$ , is heated with aqueous sodium hydroxide?

- [A] Evolution of a pungent alkaline gas (ammonia,  $\text{NH}_3$ ) that turns damp red litmus paper blue, leaving a solution of sodium ethanoate.
- [B] Vigorous effervescence of carbon dioxide that turns limewater milky.
- [C] A white precipitate of silver chloride forms.
- [D] A sweet fruity aroma of an ester.

59 [MEDIUM] Synthesis of Amides from Acyl Chlorides — 9701/11/O/N/23/Q33

[1]

What reaction conditions and stoichiometry are used to prepare ethanamide from ethanoyl chloride and ammonia?

- [A] Heat equimolar amounts under reflux with concentrated sulfuric acid.
- [B] React in the gas phase at  $500^\circ\text{C}$  over a nickel catalyst.
- [C] Add sodium borohydride to a mixture of ethanoyl chloride and nitrogen gas.
- [D] Add concentrated aqueous ammonia to ethanoyl chloride at room temperature using a 2:1 mole ratio of  $\text{NH}_3$  to  $\text{CH}_3\text{COCl}$ :  $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ .

60 [HARD] Reduction of Nitriles to Primary Amines — 9701/11/M/J/220/Q60

[1]

Which reducing agent cleanly converts ethanenitrile into ethylamine?

- [A] Sodium borohydride ( $\text{NaBH}_4$ ) in water
- [B] Lithium aluminium hydride ( $\text{LiAlH}_4$ ) in dry ether (or  $\text{H}_2$  with Ni catalyst at  $150^\circ\text{C}$ )
- [C] Acidified potassium dichromate(VI)
- [D] Concentrated sulfuric acid

**61** [EASY] Alkaline Hydrolysis of Nitriles — 9701/12/M/J/221/Q61**[1]**

What products are formed when butanenitrile is heated with aqueous sodium hydroxide?

- [A] Butanoic acid and sodium chloride
- [B] Sodium butanoate ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$ ) and ammonia gas ( $\text{NH}_3$ )
- [C] Butan-1-amine
- [D] Butanal

**62** [HARD] Carbon Chain Extension via Nitriles — 9701/13/M/J/222/Q62**[1]**

Starting from bromoethane, how can propanoic acid be synthesized in a two-step sequence?

- [A] Step 1:  $\text{NaOH(aq)}$ , reflux; Step 2: acidified potassium dichromate.
- [B] Step 1:  $\text{NH}_3$  in ethanol; Step 2:  $\text{HNO}_3$ .
- [C] Step 1:  $\text{Mg}$  in ether; Step 2: water.
- [D] Step 1:  $\text{KCN}$  in ethanol, heated under reflux; Step 2: Dilute hydrochloric acid, heated under reflux.

**63** [EASY] Cyanohydrin Hydrolysis to Alpha-Hydroxy Acids — 9701/11/M/J/223/Q63**[1]**

What reaction converts 2-hydroxypropanenitrile into 2-hydroxypropanoic acid (lactic acid)?

- [A] Heating under reflux with dilute hydrochloric acid (acid hydrolysis of the  $-\text{CN}$  group)
- [B] Reduction with  $\text{NaBH}_4$
- [C] Dehydration with concentrated sulfuric acid
- [D] Oxidation with potassium manganate(VII)

**64** [HARD] Planar Geometry of Amide Group — 9701/12/M/J/224/Q64**[1]**

Why are the atoms of the amide group ( $-\text{CO}-\text{NH}-$ ) strictly coplanar, with an  $\text{H}-\text{N}-\text{C}$  bond angle of approximately  $120^\circ$ ?

- [A] The nitrogen atom is  $\text{sp}^3$  hybridised with rapid inversion.
- [B] Amides contain triple bonds.
- [C] Resonance delocalisation gives the  $\text{C}-\text{N}$  bond partial double-bond character (bond order  $\sim 1.5$ ), forcing the nitrogen atom to adopt  $\text{sp}^2$  hybridisation.
- [D] Hydrogen bonding locks atoms into a circle.

**65** [EASY] Acid Hydrolysis of Amides — 9701/13/M/J/220/Q65**[1]**

What products are formed when ethanamide ( $\text{CH}_3\text{CONH}_2$ ) is heated under reflux with dilute sulfuric acid?

- [A] Ethanoic acid ( $\text{CH}_3\text{COOH}$ ) and ammonium sulfate,  $(\text{NH}_4)_2\text{SO}_4$
- [B] Ethylamine and sulfur dioxide
- [C] Ethanenitrile and water
- [D] Sodium ethanoate and ammonia

**66** [HARD] Dehydration of Amides to Nitriles — 9701/11/M/J/221/Q66**[1]**

What product is formed when ethanamide is heated with phosphorus(V) oxide,  $\text{P}_4\text{O}_{10}$ ?

- [A] Ethanoic acid
- [B] Ethylamine
- [C] Methane and  $\text{CO}_2$
- [D] Ethanenitrile,  $\text{CH}_3\text{C}\equiv\text{N}$ , and phosphoric acid

67 [EASY] Preparation of Substituted Amides — 9701/12/M/J/222/Q67

[1]

What organic product is formed when ethanoyl chloride reacts with excess ethylamine?

- [A] Ethanamide and chloroethane
- [B] N-ethylethanamide,  $\text{CH}_3\text{CONHCH}_2\text{CH}_3$  (a secondary amide)
- [C] Diethylamine
- [D] Ethanenitrile

68 [HARD] Polyamide Synthesis: Nylon-6,6 — 9701/13/M/J/223/Q68

[1]

What are the two monomers used in the condensation polymerisation to manufacture Nylon-6,6?

- [A] Ethene and propene
- [B] Terephthalic acid and ethylene glycol
- [C] Hexane-1,6-dioic acid (adipic acid) and 1,6-diaminohexane (hexamethylenediamine)
- [D] Phenol and formaldehyde

69 [EASY] Polyamide Synthesis: Kevlar — 9701/11/M/J/224/Q69

[1]

What are the monomers of the high-strength aromatic polyamide Kevlar?

- [A] Benzene and nylon
- [B] Benzene-1,4-dicarboxylic acid (terephthalic acid) and benzene-1,4-diamine (1,4-phenylenediamine)
- [C] Phenol and aniline
- [D] Benzoic acid and ammonia

70 [HARD] Peptide Linkage in Proteins — 9701/12/M/J/220/Q70

[1]

What is the name of the functional group that links amino acid residues together in peptides and proteins?

- [A] The peptide bond (secondary amide linkage,  $-\text{CO}-\text{NH}-$ )
- [B] The ester linkage
- [C] The ether linkage
- [D] The glycosidic bond

71 [EASY] Boiling Point of Primary Amides — 9701/13/M/J/221/Q71

[1]

Why does ethanamide ( $\text{CH}_3\text{CONH}_2$ ,  $M_r = 59.0$ ) have an extraordinarily high melting point of  $82^\circ\text{C}$  and boiling point of  $222^\circ\text{C}$  compared to ethanoic acid (bp  $118^\circ\text{C}$ )?

- [A] Ethanamide is an ionic solid of sodium.
- [B] Ethanamide contains double bonds between carbon and carbon.
- [C] Ethanamide has a much larger electron cloud.
- [D] Each primary amide group can donate two hydrogen bonds (from  $-\text{NH}_2$ ) and accept two hydrogen bonds (at  $\text{C}=\text{O}$ ), creating an extensive, dense three-dimensional hydrogen-bonded lattice.

72 [HARD] Solubility of Amides in Water — 9701/11/M/J/222/Q72

[1]

Why are lower amides (such as methanamide and ethanamide) completely miscible with water?

- [A] They react vigorously with water to form explosive gases.
- [B] They are non-polar liquids.
- [C] They form extensive hydrogen bonds with water molecules using both the carbonyl oxygen and the amide amino group.
- [D] They ionize completely into mobile ions.

**73** [EASY] Infrared Spectrum of Primary Amides — 9701/12/M/J/223/Q73**[1]**

What characteristic bands in an infrared spectrum identify a primary amide?

- [A] Two N-H stretching peaks at  $3180\text{--}3350\text{ cm}^{-1}$  and a strong carbonyl (Amide I) absorption at  $1630\text{--}1690\text{ cm}^{-1}$ .
- [B] A single sharp peak at  $2250\text{ cm}^{-1}$ .
- [C] A broad band from  $2500$  to  $3300\text{ cm}^{-1}$ .
- [D] No peaks above  $1500\text{ cm}^{-1}$ .

**74** [HARD] Infrared Spectrum of Nitriles — 9701/13/M/J/224/Q74**[1]**

What distinctive absorption band confirms the presence of a nitrile group in an organic molecule?

- [A] A broad band at  $3300\text{ cm}^{-1}$ .
- [B] A strong carbonyl peak at  $1710\text{ cm}^{-1}$ .
- [C] A sharp, medium-intensity absorption band at  $2220\text{--}2260\text{ cm}^{-1}$  due to the C#N triple bond stretch.
- [D] A fingerprint peak at  $800\text{ cm}^{-1}$ .

**75** [EASY] Hofmann Bromamide Degradation Overview — 9701/11/M/J/220/Q75**[1]**

What transformation occurs when a primary amide is heated with bromine and aqueous sodium hydroxide (Hofmann degradation)?

- [A] The amide is dehydrated to a nitrile.
- [B] The amide is converted into a primary amine with ONE FEWER carbon atom, releasing  $\text{CO}_2$ /carbonate.
- [C] The amide is chlorinated.
- [D] The amide polymerises into nylon.

**76** [HARD] Reaction of Acyl Chlorides with Secondary Amines — 9701/12/M/J/221/Q76**[1]**What product is formed when ethanoyl chloride reacts with dimethylamine,  $(\text{CH}_3)_2\text{NH}$ ?

- [A] Ethanamide and chloromethane
- [B] Trimethylamine
- [C] Ethanenitrile
- [D] N,N-dimethylethanamide,  $\text{CH}_3\text{CON}(\text{CH}_3)_2$  (a tertiary amide)

**77** [EASY] Physical State of Amides — 9701/13/M/J/222/Q77**[1]**What are the physical states of methanamide and ethanamide at room temperature ( $20^\circ\text{C}$ )?

- [A] Both are gases.
- [B] Both are volatile liquids.
- [C] Methanamide is a liquid (mp  $2^\circ\text{C}$ ); Ethanamide is a crystalline solid (mp  $82^\circ\text{C}$ ).
- [D] Both are amorphous polymers.

**78** [HARD] Hydrolysis of Nylon-6,6 — 9701/11/M/J/223/Q78**[1]**

What conditions are required to hydrolyze Nylon-6,6 chemically in the laboratory?

- [A] Prolonged heating under reflux with hot concentrated hydrochloric acid ( $6\text{ mol dm}^{-3}$ ) or hot concentrated NaOH.
- [B] Shaking with cold water for 5 minutes.
- [C] Exposing to sunlight.
- [D] Dissolving in ethanol.

**79** [EASY] Polyester vs Polyamide Biodegradability — 9701/12/M/J/224/Q79**[1]**

Why are condensation polymers (polyesters and polyamides) biodegradable, unlike addition polymers (polyethene)?

- [A] They dissolve in cold rainwater.
- [B] They sublime into the atmosphere.
- [C] They react with nitrogen gas.
- [D] Their polar ester and amide linkages can be cleaved by hydrolysis catalyzed by soil moisture, acids, or bacterial enzymes.

**80 [HARD]** Reaction of Nitriles with Grignard Reagents — 9701/13/M/J/220/Q80**[1]**

What product is formed when ethanenitrile reacts with methylmagnesium bromide followed by acid hydrolysis?

- [A] Ethanoic acid
- [B] Propanone,  $\text{CH}_3\text{COCH}_3$  (a ketone)
- [C] Propan-1-amine
- [D] Ethanol

**81 [EASY]** Nitrile Addition vs Elimination Equilibrium — 9701/11/M/J/221/Q81**[1]**

Why is the cyano group ( $-\text{C}\equiv\text{N}$ ) linear with an  $\text{sp}$  hybridised carbon and nitrogen?

- [A] The carbon is  $\text{sp}^3$  hybridised.
- [B] The carbon atom forms two  $\text{sp}$  hybrid orbitals at  $180^\circ$  (one sigma to R, one sigma to N) and two perpendicular pi bonds.
- [C] The nitrogen is  $\text{sp}^2$  hybridised with bent geometry.
- [D] The bond angle is  $109.5^\circ$ .

**82 [HARD]** Toxicity of Nitriles vs Cyanide Salts — 9701/12/M/J/222/Q82**[1]**

Why are organic nitriles (such as ethanenitrile) less acutely toxic than inorganic cyanide salts (such as KCN)?

- [A] Nitriles are completely non-toxic vitamins.
- [B] Nitriles contain no cyanide groups.
- [C] Nitriles are solid minerals.
- [D] The cyano group in nitriles is covalently bonded to carbon and does not dissociate immediately in water to release free cyanide ions ( $:\text{CN}^-$ ).

**83 [EASY]** Synthesis of Acrylonitrile Monomer — 9701/13/M/J/223/Q83**[1]**

What is acrylonitrile (propenenitrile,  $\text{CH}_2=\text{CH}-\text{C}\equiv\text{N}$ ) used for industrially, and what type of polymer does it form?

- [A] It undergoes addition polymerisation to form poly(acrylonitrile) (acrylic fibers used in synthetic wool, carpets, and blankets).
- [B] It forms polyester plastics.
- [C] It is used as a motor fuel.
- [D] It forms nylon ropes.

**84 [HARD]** Preparation of Ethanitrile from Haloalkanes — 9701/11/M/J/224/Q84**[1]**

What are the reagents and conditions to produce ethanenitrile from chloromethane?

- [A] Hydrogen cyanide gas at  $-50^\circ\text{C}$ .
- [B] Concentrated nitric acid.
- [C] Potassium cyanide (KCN) dissolved in ethanol/water, heated under reflux.
- [D] Ammonia in ethanol.

**85 [EASY]** Reduction of Amides with  $\text{LiAlH}_4$  — 9701/12/M/J/220/Q85**[1]**

What organic product is formed when ethanamide ( $\text{CH}_3\text{CONH}_2$ ) is reduced with  $\text{LiAlH}_4$  in dry ether?

- [A] Ethylamine,  $\text{CH}_3\text{CH}_2\text{NH}_2$  (the carbonyl oxygen is completely removed and replaced by two hydrogens).
- [B] Ethanol
- [C] Ethanoic acid
- [D] Ethanal

**86** [HARD] Reduction of Substituted Amides with  $\text{LiAlH}_4$  — 9701/13/M/J/221/Q86

[1]

What product is formed when N-methylethanamide,  $\text{CH}_3\text{CONHCH}_3$ , is reduced with  $\text{LiAlH}_4$ ?

- [A] Ethylamine and methane
- [B] Ethanamide
- [C] Dimethylamine
- [D] Ethylmethanamine,  $\text{CH}_3\text{CH}_2\text{NHCH}_3$  (a secondary amine)

**87** [EASY] Identification of Amide vs Nitrile by Chemical Tests — 9701/11/M/J/222/Q87

[1]

How can ethanamide and ethanenitrile be distinguished by their reactions with aqueous sodium hydroxide?

- [A] Ethanenitrile forms a silver mirror.
- [B] Ethanenitrile requires prolonged boiling to release ammonia, whereas ethanamide hydrolyzes smoothly upon warming to evolve ammonia gas.
- [C] Ethanamide effervesces with sodium carbonate.
- [D] Neither reacts with sodium hydroxide.

**88** [HARD] Amide Melting Point Trend with Alkyl Substitution — 9701/12/M/J/223/Q88

[1]

Why does N,N-dimethylethanamide,  $\text{CH}_3\text{CON}(\text{CH}_3)_2$ , have a much lower melting point ( $-20^\circ\text{C}$ , a liquid) than ethanamide,  $\text{CH}_3\text{CONH}_2$  (mp  $82^\circ\text{C}$ , a crystalline solid)?

- [A] It has a lower molecular mass.
- [B] Dimethyl substitution breaks the  $\text{C}=\text{O}$  bond.
- [C] Replacing both N-H hydrogens with methyl groups completely eliminates intermolecular hydrogen bonding between amide molecules.
- [D] It forms ionic bonds.

**89** [EASY] Industrial Solvent: Dimethylformamide (DMF) — 9701/13/M/J/224/Q89

[1]

What is N,N-dimethylmethanamide (dimethylformamide, DMF,  $\text{HCON}(\text{CH}_3)_2$ ), and why is it a widely used industrial solvent?

- [A] It is a solid wax.
- [B] It is a polar aprotic liquid with a high dielectric constant that dissolves both polar and non-polar compounds without donating hydrogen bonds.
- [C] It is an inorganic mineral acid.
- [D] It is a volatile refrigerant gas.

**90** [HARD] Cyanide Addition Regiochemistry in Conjugated Carbonyls — 9701/11/M/J/220/Q90

[1]

When an  $\alpha,\beta$ -unsaturated ketone reacts with  $\text{HCN}$ , where does the cyanide nucleophile attack?

- [A] It can undergo 1,2-addition at the carbonyl carbon OR 1,4-conjugate addition across the  $\text{C}=\text{C}$  double bond.
- [B] Only on the alkyl methyl group.
- [C] Only on oxygen.
- [D] It cleaves the molecule into two fragments.

**91** [EASY] Reaction of Nitriles with Alcohols and Acid (Pinner Reaction) — 9701/12/M/J/221/Q91

[1]

What intermediate is formed when a nitrile is treated with anhydrous alcohol and dry  $\text{HCl}$  gas?

- [A] An alkane.
- [B] An acyl chloride.
- [C] An aldehyde.
- [D] An imidate ester hydrochloride salt,  $[\text{R}-\text{C}(=\text{NH}_2^+)(\text{OR}')] \text{Cl}^-$ , which hydrolyzes to an ester.

**92** [HARD] Natural Occurrence of Nitriles: Amygdalin — 9701/13/M/J/222/Q92

[1]

What naturally occurring cyanogenic glycoside found in apricot kernels and bitter almonds releases toxic HCN upon enzymatic hydrolysis?

- [A] Chlorophyll
- [B] Hemoglobin
- [C] Amygdalin
- [D] Cellulose

**93** [EASY] Preparation of Aromatic Nitriles via Sandmeyer Reaction — 9701/11/M/J/223/Q93

[1]

How is benzonitrile ( $\text{C}_6\text{H}_5\text{CN}$ ) prepared from benzenediazonium chloride?

- [A] By warming benzenediazonium chloride with copper(I) cyanide ( $\text{CuCN}$ ) and potassium cyanide.
- [B] By reacting benzene with KCN in ethanol.
- [C] By heating chlorobenzene with ammonia.
- [D] By reducing nitrobenzene.

**94** [HARD] Deducing Monomer of Poly(hexamide) (Nylon-6) — 9701/12/M/J/224/Q94

[1]

What single cyclic monomer is polymerised via ring-opening polymerisation to produce Nylon-6?

- [A] Adipic acid
- [B] Ethene
- [C] Caprolactam (a 7-membered cyclic amide)
- [D] Hexamethylenediamine

**95** [EASY] Hydrogen Bonding in Nylon-6,6 Fibers — 9701/13/M/J/220/Q95

[1]

What intermolecular force is primarily responsible for the exceptional tensile strength and high melting point ( $265^\circ\text{C}$ ) of Nylon-6,6 fibers?

- [A] Weak London dispersion forces.
- [B] Extensive, highly regular intermolecular hydrogen bonds between the  $\text{C}=\text{O}$  of one chain and the  $\text{N}-\text{H}$  of adjacent parallel chains.
- [C] Covalent crosslinks of sulfur.
- [D] Electrostatic repulsion between carbon chains.

**96** [HARD] Quantitative Hydrolysis of Nitrile to Carboxylic Acid — 9701/11/M/J/221/Q96

[1]

0.020 mol of an unknown aliphatic nitrile is completely hydrolyzed with dilute HCl. The resulting carboxylic acid is titrated with 0.100 mol  $\text{dm}^{-3}$  NaOH.

What volume of NaOH solution is required to neutralize the carboxylic acid?

- [A] 100.0  $\text{cm}^3$  of NaOH
- [B] 50.0  $\text{cm}^3$  of NaOH
- [C] 400.0  $\text{cm}^3$  of NaOH
- [D] 200.0  $\text{cm}^3$  of NaOH

**97** [EASY] Amide Stability Towards Oxidation — 9701/12/M/J/222/Q97

[1]

How does ethanamide behave when heated with acidified potassium dichromate(VI)?

- [A] The solution turns green.
- [B] A brick-red precipitate forms.
- [C] No reaction occurs; the solution remains orange.
- [D] A silver mirror forms.

**98** [HARD] Synthesis of Acrylamide — 9701/13/M/J/223/Q98

[1]

How is acrylamide ( $\text{CH}_2=\text{CH}-\text{CONH}_2$ ), the monomer for polyacrylamide gel electrophoresis (PAGE), prepared industrially from acrylonitrile?

- [A] Controlled catalytic hydration of the nitrile group in acrylonitrile using copper catalyst:  $\text{CH}_2=\text{CH}-\text{CN} + \text{H}_2\text{O} \rightarrow \text{CH}_2=\text{CH}-\text{CONH}_2$ .
- [B] Reduction with hydrogen.
- [C] Oxidation with  $\text{KMnO}_4$ .
- [D] Esterification with methanol.

**99** [EASY] Distinction Between Amide and Amine by Odor and pH — 9701/11/M/J/224/Q99

[1]

How can ethanamide and ethylamine be distinguished immediately without chemical reagents?

- [A] Ethylamine is an odorless solid; Ethanamide is a foul-smelling gas.
- [B] Both are acidic liquids.
- [C] Both are identical in physical properties.
- [D] Ethylamine is a volatile alkaline liquid with a pungent fishy odor; Ethanamide is an odorless, neutral crystalline solid.

**100** [HARD] Catalytic Hydrogenation of Nitriles in Industry — 9701/12/M/J/220/Q100

[1]

What reagents and conditions are used in industrial manufacturing to convert propanenitrile into propan-1-amine continuously?

- [A] Concentrated sulfuric acid at  $180^\circ\text{C}$ .
- [B] Hydrogen gas ( $\text{H}_2$ ) in the presence of a finely divided nickel (Ni) or platinum (Pt) catalyst at elevated temperature and pressure.
- [C] Sodium hydroxide in ethanol under reflux.
- [D] Oxygen gas at  $500^\circ\text{C}$ .

**FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)****101** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 1: Basicity Trend Across Amine Classes — 9701/11/M/J/22/Q32

[1]

Consider the following four nitrogen-containing compounds:

1: Ammonia ( $\text{NH}_3$ ); 2: Ethylamine ( $\text{CH}_3\text{CH}_2\text{NH}_2$ ); 3: Diethylamine ( $(\text{CH}_3\text{CH}_2)_2\text{NH}$ ); 4: Phenylamine ( $\text{C}_6\text{H}_5\text{NH}_2$ )

Which sequence correctly ranks these compounds in order of INCREASING base strength (weakest base to strongest base in aqueous solution)?

- [A]  $4 < 2 < 3 < 1$
- [B]  $1 < 4 < 2 < 3$
- [C]  $4 < 1 < 2 < 3$
- [D]  $3 < 2 < 1 < 4$

**102** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 2: Synthetic Route for Extending Carbon Chain and Forming Amines — 9701/12/M/J/21/Q32

[1]

A chemist wishes to convert bromoethane into propan-1-amine.

Which reaction pathway will achieve this synthesis in the highest yield with zero formation of secondary or tertiary amines?

- [A] Reflux with KCN in ethanol to form propanenitrile, followed by reduction with  $\text{LiAlH}_4$  in dry ether.
- [B] Heat under pressure with excess concentrated ethanolic ammonia.
- [C] Reflux with aqueous sodium hydroxide followed by reaction with ammonia.
- [D] Heat with potassium cyanide followed by acid hydrolysis.

**103** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 3: Explaining the Lack of Basicity in Amides — 9701/13/O/N/22/Q32

[1]

Why is ethanamide ( $\text{CH}_3\text{CONH}_2$ ) a neutral compound in aqueous solution that fails to react with dilute hydrochloric acid at room temperature?

- [A] Ethanamide has no non-bonding electron lone pairs.
- [B] The nitrogen atom is  $\text{sp}^3$  hybridised with tetrahedral bond angles.
- [C] Ethanamide forms an explosive complex with hydrochloric acid.
- [D] The lone pair of electrons on the nitrogen atom is delocalised into the adjacent carbonyl group by resonance, drastically reducing its ability to coordinate with a proton.

**104** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 4: Identifying Reaction Products in Amide Alkaline Hydrolysis — 9701/11/F/M/22/Q32**[1]**

An unknown organic solid W is heated with aqueous sodium hydroxide.

A choking gas is evolved that turns damp red litmus paper blue. When the remaining reaction mixture is acidified with dilute sulfuric acid, a liquid X is formed that reacts with sodium carbonate to produce effervescence of CO<sub>2</sub>.

What is the identity of compound W?

- [A] Ethylamine, CH<sub>3</sub>CH<sub>2</sub>NH<sub>2</sub>
- [B] Ethanamide, CH<sub>3</sub>CONH<sub>2</sub>
- [C] Ethanenitrile, CH<sub>3</sub>CN
- [D] Aminoethanoic acid, H<sub>2</sub>NCH<sub>2</sub>COOH

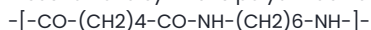
**105** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 5: Stoichiometry of Amide Synthesis from Acyl Chlorides — 9701/12/O/N/21/Q32**[1]**

How many moles of ammonia are required to react completely with 1.00 mol of ethanoyl chloride, CH<sub>3</sub>COCl, to form ethanamide, and what is the reason for this stoichiometry?

- [A] 2.00 moles of ammonia; 1.00 mol acts as the nucleophile to form ethanamide and 1.00 mol acts as a base to neutralize the liberated HCl byproduct as NH<sub>4</sub>Cl.
- [B] 1.00 mole of ammonia; ammonia acts only as a catalyst.
- [C] 3.00 moles of ammonia; each hydrogen on ammonia is substituted.
- [D] 0.50 mole of ammonia.

**106** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 6: Deducing Monomers from Polyamide Repeat Unit — 9701/13/M/J/22/Q32**[1]**

A section of a synthetic polyamide fiber has the repeating unit:



What are the systematic IUPAC names of the two monomers used to synthesize this polymer?

- [A] Butanedioic acid and 1,4-diaminobutane
- [B] Hexanoic acid and hexylamine
- [C] Hexanedioic acid and hexane-1,6-diamine
- [D] Pentanedioic acid and 1,5-diaminopentane

**107** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 7: Quantitative Nitrile Reduction Stoichiometry — 9701/11/O/N/21/Q32**[1]**

What mass of propan-1-amine (Mr = 59.1) is produced by the complete reduction of 5.50 g of propanenitrile (Mr = 55.1) using an excess of LiAlH<sub>4</sub>, assuming an 80.0% yield?

- [A] 5.90 g of propan-1-amine
- [B] 4.72 g of propan-1-amine
- [C] 3.54 g of propan-1-amine
- [D] 5.50 g of propan-1-amine

**108** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 8: Distinguishing Primary Amines from Amides by Litmus and Reagents — 9701/12/F/M/23/Q32**[1]**

Two unlabelled white crystalline solids, P and Q, are known to be ethylammonium chloride (CH<sub>3</sub>CH<sub>2</sub>NH<sub>3</sub><sup>+</sup> Cl<sup>-</sup>) and ethanamide (CH<sub>3</sub>CONH<sub>2</sub>).

Which single test immediately differentiates between solid P and solid Q?

- [A] Dissolve in water: solid P is insoluble whereas solid Q dissolves.
- [B] Add silver nitrate solution: neither forms a precipitate.
- [C] Test with universal indicator: both are strongly basic.
- [D] Add aqueous sodium hydroxide at room temperature: solid P immediately liberates volatile ethylamine gas with a fishy smell, while solid Q does not react until heated strongly.

**109** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 9: Predicting Products of Hydrolysis of Cyano-Esters — 9701/13/M/J/23/Q32**[1]**

An organic molecule contains both an ester group and a nitrile group:

$\text{NC-CH}_2\text{-COOCH}_2\text{CH}_3$  (ethyl cyanoethanoate)

What organic products are obtained upon complete acid-catalyzed hydrolysis with excess dilute hydrochloric acid under reflux?

- [A] Propanedioic acid (malonic acid,  $\text{HOOC-CH}_2\text{-COOH}$ ), ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ ), and ammonium chloride ( $\text{NH}_4\text{Cl}$ )
- [B] Aminoethanoic acid and ethanol
- [C] Ethane-1,2-diol and methanoic acid
- [D] Propanenitrile and ethanoic acid

**110** [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 10: Environmental Degradation of Poly(acrylonitrile) vs Polyethene — 9701/11/M/J/23/Q33**[1]**

Why is poly(acrylonitrile) (acrylic fiber,  $-\text{CH}_2\text{-CH}(\text{CN})-\text{n}$ ) resistant to chemical biodegradation in landfills, similar to polyethene?

- [A] It contains ionic bonds that repel water.
- [B] It is completely water-soluble and evaporates.
- [C] It is an addition polymer with a continuous, unreactive all-carbon backbone (C-C single bonds) that cannot be cleaved by hydrolysis.
- [D] It decomposes into toxic cyanide gas that kills all bacteria.

### QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

| Q1 – Q20 |   | Q21 – Q40 |   | Q41 – Q60 |   | Q61 – Q80 |   | Q81 – Q100 |   |
|----------|---|-----------|---|-----------|---|-----------|---|------------|---|
| Q01      | B | Q21       | B | Q41       | B | Q61       | B | Q81        | B |
| Q02      | D | Q22       | D | Q42       | D | Q62       | D | Q82        | D |
| Q03      | A | Q23       | A | Q43       | A | Q63       | A | Q83        | A |
| Q04      | C | Q24       | C | Q44       | C | Q64       | C | Q84        | C |
| Q05      | A | Q25       | A | Q45       | A | Q65       | A | Q85        | A |
| Q06      | D | Q26       | D | Q46       | D | Q66       | D | Q86        | D |
| Q07      | B | Q27       | B | Q47       | B | Q67       | B | Q87        | B |
| Q08      | C | Q28       | C | Q48       | C | Q68       | C | Q88        | C |
| Q09      | B | Q29       | B | Q49       | B | Q69       | B | Q89        | B |
| Q10      | A | Q30       | A | Q50       | A | Q70       | A | Q90        | A |
| Q11      | D | Q31       | D | Q51       | D | Q71       | D | Q91        | D |
| Q12      | C | Q32       | C | Q52       | C | Q72       | C | Q92        | C |
| Q13      | A | Q33       | A | Q53       | A | Q73       | A | Q93        | A |
| Q14      | C | Q34       | C | Q54       | C | Q74       | C | Q94        | C |
| Q15      | B | Q35       | B | Q55       | B | Q75       | B | Q95        | B |
| Q16      | D | Q36       | D | Q56       | D | Q76       | D | Q96        | D |
| Q17      | C | Q37       | C | Q57       | C | Q77       | C | Q97        | C |
| Q18      | A | Q38       | A | Q58       | A | Q78       | A | Q98        | A |
| Q19      | D | Q39       | D | Q59       | D | Q79       | D | Q99        | D |
| Q20      | B | Q40       | B | Q60       | B | Q80       | B | Q100       | B |

### FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

|           |           |           |           |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Q101<br>C | Q102<br>A | Q103<br>D | Q104<br>B | Q105<br>A | Q106<br>C | Q107<br>B | Q108<br>D | Q109<br>A | Q110<br>C |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|

COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MARKS |
|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1  | B   | Option B is correct. Basicity depends on the availability of the nitrogen lone pair to accept a proton: (1) In ethylamine, the electron-donating (+I) inductive effect of the ethyl group pushes electron density onto nitrogen, increasing lone-pair availability and making it stronger than ammonia; (2) In ammonia, there are no inductive effects; (3) In phenylamine, the nitrogen lone pair delocalises into the aromatic benzene pi system, drastically reducing electron density on nitrogen and making it an extremely weak base. | [1]   |
| 2  | D   | Option D is correct. Both molecules form intermolecular hydrogen bonds. However, because nitrogen is less electronegative than oxygen (electronegativities: N = 3.0, O = 3.5), the dipole in the N-H bond (N delta- - H delta+) is significantly smaller than in the O-H bond. Consequently, the N-H ... N hydrogen bond (~13 kJ/mol) is much weaker than the O-H ... O hydrogen bond (~21 kJ/mol), so ethylamine boils at a much lower temperature.                                                                                        | [1]   |
| 3  | A   | Option A is correct. When haloalkanes react with ammonia, the primary amine product possesses an active nucleophilic lone pair and readily undergoes further alkylation to yield a complex mixture of 1°, 2°, 3° amines and quaternary ammonium salts that are tedious to separate. In contrast, reduction of propanenitrile ( $\text{CH}_3\text{CH}_2\text{CN} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ ) yields propan-1-amine exclusively in high purity.                                                 | [1]   |
| 4  | C   | Option C is correct. Ethylamine is a Bronsted-Lowry base; its nitrogen lone pair coordinates with a proton from gaseous HCl in an acid-base neutralization: $\text{C}_2\text{H}_5\text{NH}_2(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{NH}_3^+ \text{Cl}^-(\text{s})$ . This produces a dense white smoke of microcrystalline solid ethylammonium chloride, which is freely soluble in water as an ionic salt.                                                                                                 | [1]   |
| 5  | A   | Option A is correct. Nitrogen has four electron pairs (three single C-H/N-H bonding pairs and one non-bonding lone pair). According to VSEPR theory, the electron pairs adopt a tetrahedral arrangement. Greater lone pair-bonding pair repulsion compresses the H-N-H and C-N-H bond angles to ~107°, yielding a trigonal pyramidal molecular geometry.                                                                                                                                                                                    | [1]   |
| 6  | D   | Option D is correct. Lower aliphatic amines form hydrogen bonds with water using both their N-H protons and the nitrogen lone pair. Furthermore, amines act as weak bases in water, accepting a proton to generate alkylammonium and hydroxide ions (pH ~11-12), giving alkaline solutions with characteristic fishy odors.                                                                                                                                                                                                                 | [1]   |
| 7  | B   | Option B is correct. Ethylamine is a stronger nucleophile than ammonia. If equimolar amounts are used, newly formed ethylamine rapidly attacks unreacted bromoethane to form diethylamine (( $\text{C}_2\text{H}_5$ ) <sub>2</sub> NH). Using a massive molar excess of NH <sub>3</sub> (e.g. 10:1) ensures that bromoethane collides almost exclusively with ammonia, yielding predominantly the primary amine.                                                                                                                            | [1]   |
| 8  | C   | Option C is correct. Ammonia has a boiling point of -33 °C and bromoethane boils at 38 °C. Heating to 100 °C to provide sufficient activation energy requires a sealed pressure tube to contain the volatile reactants in the liquid phase.                                                                                                                                                                                                                                                                                                 | [1]   |
| 9  | B   | Option B is correct. Two methyl groups exert a combined +I inductive effect, increasing electron density on nitrogen and stabilizing the protonated dialkylammonium cation (( $\text{CH}_3$ ) <sub>2</sub> NH <sub>2</sub> <sup>+</sup> ) more than in a primary amine.                                                                                                                                                                                                                                                                     | [1]   |
| 10 | A   | Option A is correct. In aqueous solution, base strength is governed by both electronic induction and hydration stabilization. Although three ethyl groups provide maximum +I donation, the bulky alkyl chains sterically shield the nitrogen lone pair and prevent water molecules from hydrogen-bonding to and stabilizing the ( $\text{C}_2\text{H}_5$ ) <sub>3</sub> NH <sup>+</sup> cation.                                                                                                                                             | [1]   |
| 11 | D   | Option D is correct. In phenylamine, the p orbital holding nitrogen's lone pair overlaps with the delocalised pi electron cloud of the benzene ring. This resonance delocalization draws electron density away from nitrogen, severely reducing its partial negative charge and making it very difficult for the lone pair to coordinate with a proton.                                                                                                                                                                                     | [1]   |
| 12 | C   | Option C is correct. In neutral phenylamine, the lone pair enjoys resonance stabilization with the ring. Upon accepting a proton to form the anilinium ion ( $\text{C}_6\text{H}_5\text{NH}_3^+$ ), nitrogen uses its lone pair to form a dative bond to H <sup>+</sup> , extinguishing resonance delocalization. This energetic penalty makes phenylamine a very weak base (it will dissolve in strong acid like HCl, but not in water).                                                                                                   | [1]   |
| 13 | A   | Option A is correct. Like ammonia, ethylamine is basic and precipitates pale blue Cu(OH) <sub>2</sub> : $\text{Cu}^{2+} + 2\text{OH}^- \rightarrow \text{Cu}(\text{OH})_2(\text{s})$ . In excess ethylamine, the amine molecules act as monodentate ligands, using their nitrogen lone pairs to coordinate with Cu <sup>2+</sup> to form a soluble, intense deep blue complex ion: $[\text{Cu}(\text{C}_2\text{H}_5\text{NH}_2)_4(\text{H}_2\text{O})_2]^{2+}$ .                                                                            | [1]   |
| 14 | C   | Option C is correct. Amines possess an unshared electron pair on nitrogen, acting as nucleophiles. Attack on the polar C-Br bond of bromoethane displaces bromide: $\text{C}_2\text{H}_5\text{NH}_2 + \text{C}_2\text{H}_5\text{Br} \rightarrow (\text{C}_2\text{H}_5)_2\text{NH}_2^+ \text{Br}^- \rightarrow [\text{base}] \rightarrow (\text{C}_2\text{H}_5)_2\text{NH}$ (diethylamine).                                                                                                                                                  | [1]   |
| 15 | B   | Option B is correct. Tetramethylammonium bromide [( $\text{CH}_3$ ) <sub>4</sub> N] <sup>+</sup> Br <sup>-</sup> is a fully alkylated quaternary ammonium salt. Nitrogen carries a permanent positive formal charge and has no lone pair or acidic hydrogens. It is an ionic solid that dissolves in water to form mobile conducting ions.                                                                                                                                                                                                  | [1]   |
| 16 | D   | Option D is correct. Wet hair and washed cotton fibers carry net negative surface charges. The positively charged quaternary nitrogen heads [( $\text{CH}_3$ ) <sub>3</sub> N-C <sub>16</sub> H <sub>33</sub> ] <sup>+</sup> bind tightly to these negative sites, coating fibers with a lubricating layer of hydrocarbon tails that eliminates static cling and softens fabrics.                                                                                                                                                           | [1]   |
| 17 | C   | Option C is correct. Primary amines form secondary amides ( $\text{RNH}_2 + \text{R}'\text{COCl} \rightarrow \text{R}'\text{CONHR} + \text{HCl}$ ); secondary amines form tertiary amides ( $\text{R}_2\text{NH} + \text{R}'\text{COCl} \rightarrow \text{R}'\text{CONR}_2 + \text{HCl}$ ). Tertiary amines ( $\text{R}_3\text{N}$ ) lack a replaceable hydrogen on nitrogen and cannot eliminate HCl to form a neutral amide bond.                                                                                                         | [1]   |
| 18 | A   | Option A is correct. Low aliphatic amines (e.g. trimethylamine) have strong, pungent fishy odors (responsible for the smell of rotting fish). Diamines such as putrescine (1,4-diaminobutane) and cadaverine (1,5-diaminopentane) are produced by bacterial decarboxylation of amino acids in decaying corpses, giving foul odors.                                                                                                                                                                                                          | [1]   |

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MARKS |
|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 19 | D   | Option D is correct. Hexane is insoluble in water and unreactive to acid. Ethylamine is basic and dissolves in dilute HCl as the ionic salt: $\text{C}_2\text{H}_5\text{NH}_2 + \text{HCl} \rightarrow \text{C}_2\text{H}_5\text{NH}_3^+ \text{Cl}^- (\text{aq})$ . Separating the aqueous layer and adding aqueous NaOH deprotonates the salt to regenerate free ethylamine: $\text{C}_2\text{H}_5\text{NH}_3^+ + \text{OH}^- \rightarrow \text{C}_2\text{H}_5\text{NH}_2 + \text{H}_2\text{O}$ .   | [1]   |
| 20 | B   | Option B is correct. Combustion equation: $4\text{CH}_3\text{NH}_2(\text{g}) + 9\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 10\text{H}_2\text{O}(\text{g}) + 2\text{N}_2(\text{g})$ . The carbon is oxidized to $\text{CO}_2$ , hydrogen to $\text{H}_2\text{O}$ , and nitrogen to stable diatomic $\text{N}_2$ gas.                                                                                                                                                                   | [1]   |
| 21 | B   | Option B is correct. Although a trivalent amine with three different groups and a lone pair is nominally chiral, the molecule rapidly flips inside out (nitrogen inversion) billions of times per second at 25 °C via a planar $\text{sp}^2$ transition state, causing spontaneous, instantaneous racemisation.                                                                                                                                                                                      | [1]   |
| 22 | D   | Option D is correct. Reduction of nitrobenzene: $\text{C}_6\text{H}_5\text{NO}_2 + 6[\text{H}] \rightarrow \text{C}_6\text{H}_5\text{NH}_2 + 2\text{H}_2\text{O}$ . Refluxing with tin metal and concentrated HCl produces the salt phenylammonium chloride, $\text{C}_6\text{H}_5\text{NH}_3^+ \text{Cl}^-$ . Adding excess NaOH deprotonates the ammonium ion to release free phenylamine (aniline), which is isolated by steam distillation.                                                      | [1]   |
| 23 | A   | Option A is correct. The $-\text{NH}_2$ group is an intensely powerful activating and 2,4,6-directing group because its lone pair delocalises strongly into the benzene ring. Electrophilic aromatic substitution is so rapid that all three activated positions (2, 4, and 6) are brominated simultaneously, yielding a dense white precipitate of 2,4,6-tribromophenylamine.                                                                                                                       | [1]   |
| 24 | C   | Option C is correct. Reaction: $\text{C}_6\text{H}_5\text{NH}_2 + \text{HNO}_2 + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{N}_2^+ \text{Cl}^- + 2\text{H}_2\text{O}$ . Benzenediazonium chloride is unstable because the $-\text{N}_2^+$ group is an extraordinary leaving group (forming inert $\text{N}_2$ gas). Above 5 °C, it hydrolyzes rapidly: $\text{C}_6\text{H}_5\text{N}_2^+ + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_5\text{OH} + \text{N}_2(\text{g}) + \text{H}^+$ . | [1]   |
| 25 | A   | Option A is correct. Cold benzenediazonium ion acts as an electrophile, attacking the 4-position of the electron-rich phenoxide ion: $\text{C}_6\text{H}_5\text{N}_2^+ + \text{C}_6\text{H}_5\text{O}^- \rightarrow \text{C}_6\text{H}_5\text{-N=N-C}_6\text{H}_4\text{-OH} + \text{Cl}^-$ . The resulting azo dye contains an extended delocalised pi system ( $-\text{N}=\text{N}-$ bridge connecting two aromatic rings) that absorbs visible light, producing vivid permanent dyes.              | [1]   |
| 26 | D   | Option D is correct. Aliphatic diazonium ions ( $\text{R-N}_2^+$ ) cannot delocalise their positive charge; they decompose instantaneously even at -20 °C via loss of $\text{N}_2$ to form carbocations ( $\text{R}^+$ ), which react with solvent. In benzenediazonium ( $\text{C}_6\text{H}_5\text{N}_2^+$ ), orbital overlap with the aromatic ring disperses the positive charge, conferring kinetic stability below 5 °C.                                                                       | [1]   |
| 27 | B   | Option B is correct. Cyclohexane has no pi electrons; its $\text{sp}^3$ carbons exert a weak +I electron donation, localizing the lone pair on nitrogen and making it a strong base. Phenylamine has an aromatic pi cloud that delocalises the lone pair away from nitrogen, destroying its basicity.                                                                                                                                                                                                | [1]   |
| 28 | C   | Option C is correct. Primary amines ( $-\text{NH}_2$ ) possess two N-H bonds that undergo coupled symmetric and asymmetric stretching vibrations, producing a characteristic doublet (two sharp peaks) in the 3300-3500 $\text{cm}^{-1}$ region. (Secondary amines have only one N-H bond and show a single peak; tertiary amines show no N-H peaks).                                                                                                                                                | [1]   |
| 29 | B   | Option B is correct. Propylamine ( $1^\circ$ ) has two N-H bonds and forms strong hydrogen bonds (bp 48 °C). Ethylmethylamine ( $2^\circ$ ) has one N-H bond and forms weaker hydrogen bonds (bp 36 °C). Trimethylamine ( $3^\circ$ ) has all hydrogens replaced by methyl groups: it has zero N-H bonds and cannot form intermolecular hydrogen bonds among its molecules, boiling at only 3 °C (a gas at room temperature).                                                                        | [1]   |
| 30 | A   | Option A is correct. As an alkaline amine, ethylamine captures atmospheric acidic $\text{CO}_2$ , forming carbamate salts that precipitate as a white crust on the neck of unsealed bottles: $2\text{RNH}_2 + \text{CO}_2 \rightarrow \text{RNH}_3^+ \text{RNHCOO}^-$ (similar to ammonia capturing $\text{CO}_2$ in the manufacture of urea).                                                                                                                                                       | [1]   |
| 31 | D   | Option D is correct. Industrial synthesis: $2\text{NH}_3 + \text{CO}_2 \rightleftharpoons \text{H}_2\text{N-COONH}_4$ (ammonium carbamate) $\rightarrow \text{H}_2\text{N-CO-NH}_2$ (urea) + $\text{H}_2\text{O}$ at 200 °C and 200 atm. Urea is the world's most widely used solid nitrogen fertilizer (~46% N) and a key raw material for melamine and urea-formaldehyde resins.                                                                                                                   | [1]   |
| 32 | C   | Option C is correct. Many drug molecules are large, lipophilic amines with poor water solubility. Reacting with HCl converts the neutral amine into a water-soluble ionic salt ( $\text{Drug-NH}^+ \text{Cl}^-$ ), allowing rapid dissolution in digestive fluids and efficient absorption into systemic circulation.                                                                                                                                                                                | [1]   |
| 33 | A   | Option A is correct. Aliphatic diazonium ions are intensely unstable and decompose instantaneously upon formation: $\text{C}_2\text{H}_5\text{NH}_2 + \text{HNO}_2 \rightarrow [\text{C}_2\text{H}_5\text{N}_2^+] + \text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{OH} + \text{N}_2(\text{g}) + \text{H}^+$ . Rapid bubbling of colorless nitrogen gas is diagnostic for primary aliphatic amines.                                                                                         | [1]   |
| 34 | C   | Option C is correct. Primary aliphatic amines instantly release $\text{N}_2$ gas (fizzing) and form alcohols. Phenylamine forms a cold, clear, stable solution of benzenediazonium chloride ( $\text{C}_6\text{H}_5\text{N}_2^+ \text{Cl}^-$ ) without gas evolution, which yields a vivid orange-red precipitate upon adding alkaline phenol.                                                                                                                                                       | [1]   |
| 35 | B   | Option B is correct. In the gas phase with no water molecules to cause steric hydration problems, the inductive effect (+I) operates purely: each additional electron-releasing methyl group increases electron density on nitrogen. Thus, the intrinsic gas-phase basicity strictly follows: $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$ .                                                                                                                                                          | [1]   |
| 36 | D   | Option D is correct. Charge dispersal increases thermodynamic stability. The electron-donating inductive effect of the alkyl group pushes electron density toward the positively charged nitrogen atom, dispersing the charge over a larger volume and shifting the base equilibrium to the right.                                                                                                                                                                                                   | [1]   |
| 37 | C   | Option C is correct. Nucleophilicity is kinetically controlled and highly sensitive to steric hindrance. Quinuclidine has its three alkyl chains tied back into a rigid bicyclic cage, leaving its nitrogen lone pair completely unhindered and making it a ferocious nucleophile, whereas flexible ethyl groups in triethylamine obstruct nucleophilic attack.                                                                                                                                      | [1]   |
| 38 | A   | Option A is correct. The azide synthesis is a clean, modern route: nucleophilic substitution by azide ion ( $:\text{N}_3^-$ ) gives alkyl azide ( $\text{R-N}_3$ ). Reduction with $\text{LiAlH}_4$ cleanly yields the primary amine ( $\text{R-NH}_2$ ) with 100% selectivity and zero secondary or tertiary amine contamination.                                                                                                                                                                   | [1]   |

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                               | MARKS |
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| 39 | D   | Option D is correct. Phthalimide is deprotonated by KOH to potassium phthalimide, which has a single nucleophilic nitrogen. Alkylation with a haloalkane gives N-alkylphthalimide, which is hydrolyzed with hydrazine to release the pure primary amine exclusively.                                                                                                                                                                                             | [1]   |
| 40 | B   | Option B is correct. Nucleophilic addition of a primary amine ( $R'NH_2$ ) to a carbonyl ( $>C=O$ ) followed by elimination of water yields an imine (Schiff base) containing a carbon-nitrogen double bond: $>C=O + H_2N-R' \rightarrow >C=N-R' + H_2O$ .                                                                                                                                                                                                       | [1]   |
| 41 | B   | Option B is correct. Reaction: $RNH_2 + CHCl_3 + 3KOH \rightarrow R-NC + 3KCl + 3H_2O$ . Only primary amines form volatile carbylamines (isocyanides), which possess an intolerably revolting, putrid smell detectable even at parts-per-billion levels.                                                                                                                                                                                                         | [1]   |
| 42 | D   | Option D is correct. A primary amine forms N-alkylbenzenesulfonamide ( $C_6H_5SO_2NHR$ ). The remaining hydrogen on nitrogen is made strongly acidic by the $SO_2$ group, dissolving in KOH as a soluble salt. Acidifying with HCl regenerates the insoluble sulfonamide precipitate, uniquely identifying a primary amine.                                                                                                                                      | [1]   |
| 43 | A   | Option A is correct. Secondary amines lack an acidic hydrogen on the resulting sulfonamide ( $C_6H_5SO_2NR_2$ ), so the solid precipitate remains insoluble in KOH. Tertiary amines lack an N-H bond and do not react at all, remaining as an insoluble oily amine layer that dissolves in dilute HCl.                                                                                                                                                           | [1]   |
| 44 | C   | Option C is correct. Oxidation of primary amines by peracids or $KMnO_4$ oxidizes the nitrogen atom sequentially through hydroxylamines ( $R-NHOH$ ) and nitroso compounds ( $R-NO$ ) to nitroalkanes ( $R-NO_2$ ).                                                                                                                                                                                                                                              | [1]   |
| 45 | A   | Option A is correct. Moles of C = $53.3/12.0 = 4.44$ ; H = $15.6/1.0 = 15.6$ ; N = $31.1/14.0 = 2.22$ . Dividing by 2.22: C: $4.44/2.22 = 2.0$ ; H: $15.6/2.22 = 7.0$ ; N: $2.22/2.22 = 1.0$ . Empirical formula is $C_2H_7N$ (formula mass = 45.0). Matches $M_r = 45$ , identifying ethylamine or dimethylamine.                                                                                                                                               | [1]   |
| 46 | D   | Option D is correct. Lower amines dissolve freely because their polar $-NH_2$ group forms hydrogen bonds with water. As the alkyl chain lengthens, the non-polar, hydrophobic hydrocarbon chain becomes increasingly dominant. Dissolving a large non-polar tail requires breaking many strong water-water hydrogen bonds without sufficient compensating solute-solvent interactions, making dissolution thermodynamically unfavorable.                         | [1]   |
| 47 | B   | Option B is correct. Because alkyl groups are electron-donating (+I), each successive alkylation produces an amine with comparable or greater nucleophilicity at nitrogen. Diethylamine formed by the initial substitution readily attacks remaining bromoethane to form triethylamine, which further attacks to form the quaternary ammonium salt.                                                                                                              | [1]   |
| 48 | C   | Option C is correct. Primary amines react smoothly with acid anhydrides via nucleophilic addition-elimination: $(CH_3CO)_2O + CH_3CH_2NH_2 \rightarrow CH_3CONHCH_2CH_3 + CH_3COOH$ . The nucleophilic nitrogen lone pair attacks one carbonyl carbon, displacing the carboxylate leaving group to yield an N-substituted secondary amide and a carboxylic acid.                                                                                                 | [1]   |
| 49 | B   | Option B is correct. Both are isomers with $M_r = 45.0$ . However, ethylamine is a primary amine with two N-H bonds capable of acting as hydrogen bond donors to adjacent molecules, allowing for a branched 3D hydrogen-bonded network. Dimethylamine is a secondary amine with only one N-H donor per molecule, resulting in fewer hydrogen bonds and a lower boiling point.                                                                                   | [1]   |
| 50 | A   | Option A is correct. Oxygen (electronegativity 3.5) holds its non-bonding lone pairs much more tightly than nitrogen (electronegativity 3.0). Consequently, the lone pair on nitrogen is at a higher energy level and is far more available to coordinate with a proton ( $H^+$ ) to form a dative covalent bond than the lone pairs on oxygen.                                                                                                                  | [1]   |
| 51 | D   | Option D is correct. Each amine group can undergo nucleophilic substitution with haloalkane ends. Intermolecular reaction followed by intramolecular ring closure yields cyclic diamines like piperazine (a 6-membered heterocyclic ring with two NH groups) or extended linear polyalkylene amines.                                                                                                                                                             | [1]   |
| 52 | C   | Option C is correct. $LiAlH_4$ is a powerful hydride donor. In ethanenitrile ( $CH_3-C\equiv N$ ), two hydride ions add to carbon and two protons to nitrogen, reducing the triple bond to a single bond ( $CH_3CH_2NH_2$ ). In ethanamide ( $CH_3-CO-NH_2$ ), hydride reduces the carbonyl group to $-CH_2-$ with loss of the oxygen atom as water/aluminate, yielding the same 2-carbon primary amine ethylamine.                                              | [1]   |
| 53 | A   | Option A is correct. Ethylammonium ion is the conjugate acid of ethylamine: $CH_3CH_2NH_3^+ + OH^- \rightarrow CH_3CH_2NH_2 + H_2O$ . Adding a strong base like NaOH removes the acidic proton from nitrogen, liberating neutral volatile ethylamine (which can be distilled off or detected by its fishy smell and turning red litmus blue).                                                                                                                    | [1]   |
| 54 | C   | Option C is correct. In non-polar solvents or the gas phase, solvent hydration does not stabilize the cation. Therefore, basicity is determined purely by the electron-donating inductive (+I) effect of alkyl groups, which increases electron density on nitrogen with each added methyl group: $3^\circ$ (three methyls) $>$ $2^\circ$ (two methyls) $>$ $1^\circ$ (one methyl) $>$ $NH_3$ (no alkyls).                                                       | [1]   |
| 55 | B   | Option B is correct. Ethylamine is a weak base, generating $OH^-$ in water which precipitates brown silver(I) oxide: $2Ag^+ + 2OH^- \rightarrow Ag_2O(s) + H_2O$ . In excess ethylamine, the neutral amine molecules act as monodentate ligands, using their nitrogen lone pairs to coordinate linearly with $Ag^+$ , dissolving the precipitate: $Ag_2O + 4CH_3CH_2NH_2 + H_2O \rightarrow 2[Ag(CH_3CH_2NH_2)_2]^+ + 2OH^-$ .                                   | [1]   |
| 56 | D   | Option D is correct. In an amide, the p orbital holding nitrogen's lone pair overlaps with the adjacent carbonyl pi system: $[CH_3-C(=O)-NH_2 \leftrightarrow CH_3-C(-O^-)=NH_2^+]$ . This strong resonance delocalization pulls electron density away from nitrogen toward the electronegative oxygen atom, giving the C-N bond partial double-bond character and preventing the lone pair from coordinating with a proton. Hence, amides are neutral in water. | [1]   |
| 57 | C   | Option C is correct. Refluxing a nitrile with dilute mineral acid hydrolyzes the carbon-nitrogen triple bond: $CH_3CH_2C\equiv N + 2H_2O + HCl \rightarrow CH_3CH_2COOH + NH_4Cl$ . The nitrile carbon is oxidized to a carboxylic acid and the nitrogen atom is converted into the ammonium cation.                                                                                                                                                             | [1]   |
| 58 | A   | Option A is correct. Heating an amide with aqueous alkali hydrolyzes the amide linkage: $CH_3CONH_2 + NaOH \rightarrow CH_3COONa + NH_3(g)$ . Ammonia gas is evolved, which has a characteristic choking pungent smell and turns damp red litmus paper blue, leaving sodium ethanoate in the flask.                                                                                                                                                              | [1]   |

| Q  | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                             | MARKS |
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| 59 | D   | Option D is correct. Acyl chlorides react vigorously with ammonia at room temperature via an addition-elimination mechanism. Two moles of ammonia are required per mole of acyl chloride: one mole nucleophilically attacks the carbonyl carbon to form the amide, and the second mole neutralizes the liberated HCl byproduct to form solid ammonium chloride: $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ .          | [1]   |
| 60 | B   | Option B is correct. Nitriles are reduced to primary amines using the strong reducing agent $\text{LiAlH}_4$ : $\text{CH}_3\text{C}\equiv\text{N} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2$ (or catalytic hydrogenation with $\text{H}_2$ and Raney nickel). $\text{NaBH}_4$ is too weak to reduce nitriles.                                                                                                                                                | [1]   |
| 61 | B   | Option B is correct. Alkaline hydrolysis converts the nitrile group into a carboxylate salt and releases gaseous ammonia: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{COONa} + \text{NH}_3(\text{g})$ . Acidification with dilute acid precipitates butanoic acid.                                                                                                                       | [1]   |
| 62 | D   | Option D is correct. Step 1 extends the chain from 2 to 3 carbons via nucleophilic substitution: $\text{CH}_3\text{CH}_2\text{Br} + \text{KCN} \rightarrow \text{CH}_3\text{CH}_2\text{CN} + \text{KBr}$ . Step 2 hydrolyzes the cyano group to a carboxylic acid: $\text{CH}_3\text{CH}_2\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{NH}_4\text{Cl}$ .                                                                 | [1]   |
| 63 | A   | Option A is correct. Refluxing the cyanohydrin with dilute aqueous HCl selectively hydrolyzes the nitrile group to a carboxylic acid without altering the secondary alcohol group: $\text{CH}_3\text{CH}(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{COOH} + \text{NH}_4\text{Cl}$ .                                                                                                                               | [1]   |
| 64 | C   | Option C is correct. Delocalization of nitrogen's lone pair into the carbonyl pi system [ $-\text{C}(=\text{O})-\text{NH}- \leftrightarrow -\text{C}(\text{O}^-)=\text{N}^+\text{H}-$ ] gives the C-N bond ~40% double-bond character, restricting rotation and enforcing planar $\text{sp}^2$ geometry ( $120^\circ$ bond angles) on both carbon and nitrogen.                                                                                                                | [1]   |
| 65 | A   | Option A is correct. Acid hydrolysis: $2\text{CH}_3\text{CONH}_2 + \text{H}_2\text{SO}_4 + 2\text{H}_2\text{O} \rightarrow 2\text{CH}_3\text{COOH} + (\text{NH}_4)_2\text{SO}_4$ . In acidic conditions, the ammonia liberated is immediately protonated to the ammonium ion ( $\text{NH}_4^+$ ), forming ammonium sulfate.                                                                                                                                                    | [1]   |
| 66 | D   | Option D is correct. Phosphorus(V) oxide is a powerful dehydrating agent. Heating an unsubstituted primary amide with $\text{P}_4\text{O}_{10}$ removes a molecule of water to yield the corresponding nitrile: $\text{CH}_3\text{CONH}_2 + \text{P}_4\text{O}_{10} \rightarrow \text{CH}_3\text{CN} + 4\text{HPO}_3$ .                                                                                                                                                        | [1]   |
| 67 | B   | Option B is correct. Nucleophilic addition-elimination of ethylamine on ethanoyl chloride: $\text{CH}_3\text{COCl} + 2\text{C}_2\text{H}_5\text{NH}_2 \rightarrow \text{CH}_3\text{CONHC}_2\text{H}_5 + \text{C}_2\text{H}_5\text{NH}_3^+ \text{Cl}^-$ . The product is an N-alkyl substituted (secondary) amide.                                                                                                                                                              | [1]   |
| 68 | C   | Option C is correct. Nylon-6,6 is a synthetic polyamide formed by the condensation of a 6-carbon dicarboxylic acid (adipic acid) and a 6-carbon diamine (hexamethylenediamine): $n \text{HOOC}-(\text{CH}_2)_4-\text{COOH} + n \text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2 \rightarrow [-\text{CO}-(\text{CH}_2)_4-\text{CONH}-(\text{CH}_2)_6-\text{NH}-]_n + (2n-1) \text{H}_2\text{O}$ .                                                                                 | [1]   |
| 69 | B   | Option B is correct. Kevlar is an aromatic polyamide (aramid) synthesized from terephthalic acid (or terephthaloyl chloride) and 1,4-phenylenediamine. Rigid aromatic rings and strong intermolecular hydrogen bonds between parallel amide chains give Kevlar extraordinary tensile strength (used in bulletproof vests).                                                                                                                                                     | [1]   |
| 70 | A   | Option A is correct. Amino acids polymerise by condensation between the alpha-amino group of one amino acid and the alpha-carboxyl group of another, forming a secondary amide linkage called a peptide bond ( $-\text{CO}-\text{NH}-$ ).                                                                                                                                                                                                                                      | [1]   |
| 71 | D   | Option D is correct. In ethanamide, the two N-H protons and the two lone pairs on the carbonyl oxygen participate in extensive multi-directional intermolecular hydrogen bonding. In the solid and liquid states, this creates a dense hydrogen-bonded network that requires immense thermal energy to disrupt.                                                                                                                                                                | [1]   |
| 72 | C   | Option C is correct. Water molecules form strong hydrogen bonds to both the C=O oxygen and the -NH <sub>2</sub> hydrogens of amides, allowing lower amides to dissolve readily in all proportions.                                                                                                                                                                                                                                                                             | [1]   |
| 73 | A   | Option A is correct. Primary amides exhibit two distinct N-H stretching bands (due to symmetric and asymmetric stretching of the -NH <sub>2</sub> group) around $3180\text{--}3350 \text{ cm}^{-1}$ , accompanied by an intense carbonyl stretch (the 'Amide I' band) at $1630\text{--}1690 \text{ cm}^{-1}$ (shifted to lower frequency than ketones due to resonance).                                                                                                       | [1]   |
| 74 | C   | Option C is correct. The carbon-nitrogen triple bond ( $\text{C}\equiv\text{N}$ ) produces a sharp, clean absorption band in the $2220\text{--}2260 \text{ cm}^{-1}$ region, a unique spectral zone where few other organic functional groups absorb (only alkynes $\text{C}\equiv\text{C}$ absorb nearby, but $\text{C}\equiv\text{N}$ is much more intense due to higher polarity).                                                                                          | [1]   |
| 75 | B   | Option B is correct. The Hofmann bromamide rearrangement ( $\text{RCONH}_2 + \text{Br}_2 + 4\text{NaOH} \rightarrow \text{RNH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$ ) degrades a primary amide to a primary amine with the loss of the carbonyl carbon, providing a classic step-down method in organic synthesis.                                                                                                                                 | [1]   |
| 76 | D   | Option D is correct. Nucleophilic substitution by dimethylamine: $\text{CH}_3\text{COCl} + 2(\text{CH}_3)_2\text{NH} \rightarrow \text{CH}_3\text{CON}(\text{CH}_3)_2 + (\text{CH}_3)_2\text{NH}_2^+ \text{Cl}^-$ . The product is a tertiary amide with no N-H bonds.                                                                                                                                                                                                         | [1]   |
| 77 | C   | Option C is correct. Methanamide ( $\text{HCONH}_2$ ) is one of the few liquid amides at room temperature (mp $2.5^\circ\text{C}$ ). All other unsubstituted primary amides (including ethanamide, mp $82^\circ\text{C}$ ) are crystalline solids at room temperature due to strong intermolecular hydrogen bonding.                                                                                                                                                           | [1]   |
| 78 | A   | Option A is correct. The amide linkages in polyamides are robust and chemically resistant. Hydrolysis requires severe conditions: boiling under reflux with $6 \text{ mol/dm}^3 \text{ HCl}$ for several hours cleaves the polymer back into hexane-1,6-dioic acid and 1,6-diaminohexane dihydrochloride.                                                                                                                                                                      | [1]   |
| 79 | D   | Option D is correct. Addition polymers (like polyethene) have inert non-polar C-C backbones that resist chemical attack. Polyesters and polyamides have polar, heteroatom-containing linkages ( $-\text{COO}-$ and $-\text{CONH}-$ ) that are vulnerable to nucleophilic attack by water (hydrolysis), enabling biodegradation by environmental microbes.                                                                                                                      | [1]   |
| 80 | B   | Option B is correct. A Grignard reagent adds across the $\text{C}\equiv\text{N}$ triple bond: $\text{CH}_3\text{C}\equiv\text{N} + \text{CH}_3\text{MgBr} \rightarrow (\text{CH}_3)_2\text{C}=\text{N}-\text{MgBr}$ . Subsequent acid hydrolysis converts the imine intermediate into a ketone: $(\text{CH}_3)_2\text{C}=\text{N}-\text{MgBr} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{COCH}_3 + \text{NH}_4\text{Cl} + \text{Mg}^{2+} + \text{Br}^-$ . | [1]   |

| Q   | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MARKS |
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| 81  | B   | Option B is correct. In a nitrile ( $R-C\equiv N$ ), the carbon and nitrogen atoms form one sigma bond and two pi bonds using mutually perpendicular 2p orbitals. The steric number on carbon is 2, giving $sp$ hybridisation and a strictly linear $R-C\equiv N$ geometry ( $180^\circ$ bond angle).                                                                                                                                                                                                                                       | [1]   |
| 82  | D   | Option D is correct. Inorganic cyanides (KCN) are ionic and instantly release free, deadly $:CN^-$ ions in aqueous body fluids. In nitriles, the C-CN bond is a strong covalent bond. Although metabolized slowly by liver cytochrome P450 enzymes to release cyanide, acute toxicity is significantly slower and lower than for free cyanide salts.                                                                                                                                                                                        | [1]   |
| 83  | A   | Option A is correct. Acrylonitrile contains an alkene $C=C$ double bond activated by the electron-withdrawing cyano group. Free radical addition polymerisation yields poly(acrylonitrile) (PAN / Orlon / Cashmilon), an acrylic polymer with high thermal resistance spun into synthetic fibers.                                                                                                                                                                                                                                           | [1]   |
| 84  | C   | Option C is correct. Chloromethane reacts with ethanolic KCN under reflux via an $SN_2$ nucleophilic substitution: $CH_3Cl + :CN^- \rightarrow CH_3CN + Cl^-$ .                                                                                                                                                                                                                                                                                                                                                                             | [1]   |
| 85  | A   | Option A is correct. Reduction of amides with $LiAlH_4$ does not cleave the C-N bond; instead, the carbonyl group ( $>C=O$ ) is completely reduced to a methylene group ( $-CH_2-$ ): $RCONH_2 + 4[H] \rightarrow RCH_2NH_2 + H_2O$ . Ethanamide yields ethylamine.                                                                                                                                                                                                                                                                         | [1]   |
| 86  | D   | Option D is correct. Reduction of secondary amides by $LiAlH_4$ converts the carbonyl to methylene: $CH_3CONHCH_3 + 4[H] \rightarrow CH_3CH_2NHCH_3$ (ethylmethylamine). This provides a clean synthetic route to secondary and tertiary amines.                                                                                                                                                                                                                                                                                            | [1]   |
| 87  | B   | Option B is correct. Ethanamide is an amide and releases ammonia gas readily upon gentle warming with aqueous NaOH: $CH_3CONH_2 + OH^- \rightarrow CH_3COO^- + NH_3(g)$ . Ethanenitrile hydrolyzes much more sluggishly, requiring prolonged boiling under reflux.                                                                                                                                                                                                                                                                          | [1]   |
| 88  | C   | Option C is correct. In $N,N$ -dimethylethanamide, there are no hydrogen atoms bonded to nitrogen. Although the $C=O$ bond is polar, the molecules cannot form intermolecular hydrogen bonds among themselves, causing its melting point to drop by over $100^\circ C$ so that it is a liquid at room temperature.                                                                                                                                                                                                                          | [1]   |
| 89  | B   | Option B is correct. DMF is a polar aprotic solvent: it has a high dipole moment and solvates cations strongly while leaving nucleophilic anions naked and highly reactive, making it an ideal solvent for $SN_2$ substitution reactions and polymer synthesis (e.g. polyacrylonitrile spinning).                                                                                                                                                                                                                                           | [1]   |
| 90  | A   | Option A is correct. Conjugated enones ( $C=C-C=O$ ) have two electrophilic centers: the carbonyl carbon (direct 1,2-addition) and the beta-carbon of the alkene (1,4-conjugate addition / Michael addition). Cyanide frequently adds conjugate to yield beta-cyano ketones.                                                                                                                                                                                                                                                                | [1]   |
| 91  | D   | Option D is correct. The Pinner synthesis: nucleophilic addition of alcohol across the protonated nitrile triple bond yields an imidate salt: $RCN + R'OH + HCl \rightarrow R-C(=NH_2^+ Cl^-)OR'$ . Adding water hydrolyzes the imidate into an ester and ammonium chloride: $RCOOR' + NH_4Cl$ .                                                                                                                                                                                                                                            | [1]   |
| 92  | C   | Option C is correct. Amygdalin is a cyanogenic glycoside found in the seeds of apricots, peaches, and bitter almonds. When chewed, plant enzymes (emulsin) hydrolyze amygdalin into gentiobiose, benzaldehyde, and toxic hydrogen cyanide (HCN).                                                                                                                                                                                                                                                                                            | [1]   |
| 93  | A   | Option A is correct. The Sandmeyer reaction: copper(I) cyanide catalyzes the nucleophilic substitution of the diazonium group by cyanide: $C_6H_5N_2^+ Cl^- + CuCN \rightarrow C_6H_5CN + N_2(g) + CuCl$ , providing an essential route to aromatic carboxylic acids via subsequent hydrolysis.                                                                                                                                                                                                                                             | [1]   |
| 94  | C   | Option C is correct. Nylon-6 is manufactured by the ring-opening polymerisation of caprolactam (a 7-membered cyclic amide / lactam). Heating with trace water cleaves the ring into 6-aminohexanoic acid, which polymerises into Nylon-6: $-[CO-(CH_2)_5-NH]-_n$ .                                                                                                                                                                                                                                                                          | [1]   |
| 95  | B   | Option B is correct. The regular spacing of 6-carbon segments allows all $C=O$ and $N-H$ groups along adjacent polymer chains to align perfectly, forming continuous sheets of strong intermolecular hydrogen bonds ( $C=O \cdots H-N$ ) that give nylon fibers immense strength and elasticity.                                                                                                                                                                                                                                            | [1]   |
| 96  | D   | Option D is correct. Hydrolysis of 1 mole of mono-nitrile yields exactly 1 mole of mono-carboxylic acid: $RCN \rightarrow RCOOH$ . Thus, 0.020 mol of nitrile produces 0.020 mol of carboxylic acid. Volume of 0.100 mol/dm <sup>3</sup> NaOH needed: $V = \text{moles} / M = 0.020 \text{ mol} / 0.100 \text{ mol/dm}^3 = 0.200 \text{ dm}^3 = 200.0 \text{ cm}^3$ .                                                                                                                                                                       | [1]   |
| 97  | C   | Option C is correct. Amides are resistant to oxidation under standard laboratory conditions. Acidified potassium dichromate cannot oxidize ethanamide, and the orange $Cr_2O_7^{2-}$ solution remains unchanged.                                                                                                                                                                                                                                                                                                                            | [1]   |
| 98  | A   | Option A is correct. Acrylonitrile is selectively hydrated across its $C\equiv N$ bond using a reduced copper catalyst at $100^\circ C$ : $CH_2=CH-CN + H_2O \rightarrow CH_2=CH-CONH_2$ . The resulting acrylamide monomer is polymerised into water-absorbent polyacrylamide hydrogels and electrophoresis gels.                                                                                                                                                                                                                          | [1]   |
| 99  | D   | Option D is correct. Ethylamine has a boiling point of $16.6^\circ C$ , turns damp red litmus paper blue, and has a strong, repulsive fishy/ammoniacal smell. Ethanamide is an odorless, neutral (pH 7) white crystalline solid with a melting point of $82^\circ C$ .                                                                                                                                                                                                                                                                      | [1]   |
| 100 | B   | Option B is correct. In large-scale industrial chemical synthesis, catalytic hydrogenation is preferred over stoichiometric $LiAlH_4$ due to lower cost and lack of hazardous hydride waste: $CH_3CH_2C\equiv N + 2H_2 \xrightarrow{Ni, 150^\circ C, \text{ pressure}} CH_3CH_2CH_2NH_2$ . The nitrile triple bond is cleanly hydrogenated across to a primary amine.                                                                                                                                                                       | [1]   |
| 101 | C   | Option C is correct. Phenylamine (4) is by far the weakest base because the nitrogen lone pair delocalises into the benzene pi cloud ( $pK_b \sim 9.4$ ). Ammonia (1) has no inductive effects ( $pK_b \sim 4.75$ ). Ethylamine (2) is stronger than ammonia due to +I electron donation from one ethyl group ( $pK_b \sim 3.35$ ). Diethylamine (3) is the strongest base because two ethyl groups provide greater +I electron release without excessive steric hindrance ( $pK_b \sim 3.0$ ). Correct increasing order: $4 < 1 < 2 < 3$ . | [1]   |

| Q   | KEY | EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MARKS |
|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 102 | A   | Option A is correct. Direct reaction of bromoethane with ammonia forms ethylamine (2 carbons), not propan-1-amine (3 carbons), and produces a mixture of amines. To lengthen the carbon chain by one carbon atom, bromoethane is reacted with ethanolic KCN under reflux to yield propanenitrile ( $\text{CH}_3\text{CH}_2\text{CN}$ ). Subsequent reduction of propanenitrile with $\text{LiAlH}_4$ in dry ether yields pure propan-1-amine ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ ) with no polyalkylation byproducts.                                                                                                                                                                                                        | [1]   |
| 103 | D   | Option D is correct. The nitrogen lone pair in an amide overlaps with the pi orbital of the adjacent $\text{C}=\text{O}$ double bond: $[\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2 \leftrightarrow \text{CH}_3-\text{C}(\text{O}^-)=\text{NH}_2^+]$ . This resonance delocalisation confers significant partial double-bond character on the $\text{C}-\text{N}$ bond and draws electron density toward the electronegative oxygen atom. The nitrogen lone pair is effectively unavailable for donation to a proton, making amides neutral.                                                                                                                                                                                              | [1]   |
| 104 | B   | Option B is correct. Heating an amide with aqueous $\text{NaOH}$ hydrolyzes it to release ammonia gas (which turns damp red litmus blue) and forms a carboxylate salt: $\text{CH}_3\text{CONH}_2 + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{NH}_3(\text{g})$ . Subsequent acidification with dilute $\text{H}_2\text{SO}_4$ converts sodium ethanoate into ethanoic acid (liquid X): $2\text{CH}_3\text{COONa} + \text{H}_2\text{SO}_4 \rightarrow 2\text{CH}_3\text{COOH} + \text{Na}_2\text{SO}_4$ . Ethanoic acid effervesces with $\text{Na}_2\text{CO}_3$ ( $2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{CO}_2 + \text{H}_2\text{O}$ ), confirming W as ethanamide. | [1]   |
| 105 | A   | Option A is correct. The overall reaction is: $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$ . The first mole of $\text{NH}_3$ undergoes nucleophilic addition-elimination with ethanoyl chloride, expelling $\text{HCl}$ . The second mole of basic $\text{NH}_3$ immediately neutralizes the acidic $\text{HCl}$ gas to form solid ammonium chloride ( $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$ ). Thus, exactly 2 moles of ammonia are consumed per mole of acyl chloride.                                                                                                                                                                                     | [1]   |
| 106 | C   | Option C is correct. Cleaving the amide bonds ( $-\text{CO}-\text{NH}-$ ) by hydrolysis: the diacid fragment is $-\text{CO}-(\text{CH}_2)_4-\text{CO}-$ , which upon adding two $-\text{OH}$ groups gives $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$ (hexanedioic acid / adipic acid, 6 carbons). The diamine fragment is $-\text{NH}-(\text{CH}_2)_6-\text{NH}-$ , which upon adding two $-\text{H}$ atoms gives $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$ (hexane-1,6-diamine / hexamethylenediamine, 6 carbons). The polymer is Nylon-6,6.                                                                                                                                                                                       | [1]   |
| 107 | B   | Option B is correct. Moles of propanenitrile = $5.50 \text{ g} / 55.1 \text{ g/mol} = 0.0998 \text{ mol}$ . The reaction has a 1 : 1 stoichiometry: $\text{CH}_3\text{CH}_2\text{CN} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ . Theoretical yield of amine = $0.0998 \text{ mol} * 59.1 \text{ g/mol} = 5.90 \text{ g}$ . At 80.0% yield: Actual mass = $5.90 \text{ g} * 0.800 = 4.72 \text{ g}$ .                                                                                                                                                                                                                                                                                                        | [1]   |
| 108 | D   | Option D is correct. Solid P is an alkylammonium salt (ethylammonium chloride). Adding aqueous $\text{NaOH}$ at room temperature immediately deprotonates the cation: $\text{C}_2\text{H}_5\text{NH}_3^+ \text{Cl}^- + \text{OH}^- \rightarrow \text{C}_2\text{H}_5\text{NH}_2(\text{g}) + \text{H}_2\text{O} + \text{Cl}^-$ , liberating volatile ethylamine with its unmistakable fishy odor. Solid Q is ethanamide: amides do not release ammonia at room temperature and require boiling under reflux to undergo alkaline hydrolysis.                                                                                                                                                                                                 | [1]   |
| 109 | A   | Option A is correct. Acid hydrolysis cleaves both hydrolyzable functional groups: (1) The nitrile group ( $-\text{C}\equiv\text{N}$ ) is hydrolyzed to a carboxylic acid ( $-\text{COOH}$ ) and an ammonium ion ( $\text{NH}_4^+$ ); (2) The ester group ( $-\text{COOCH}_2\text{CH}_3$ ) is hydrolyzed to a carboxylic acid ( $-\text{COOH}$ ) and ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ ). Thus, the central carbon framework becomes the dicarboxylic acid propanedioic acid (malonic acid, $\text{HOOC}-\text{CH}_2-\text{COOH}$ ), accompanied by ethanol and ammonium chloride.                                                                                                                                               | [1]   |
| 110 | C   | Option C is correct. Poly(acrylonitrile) is formed by addition polymerisation across the alkene double bond of $\text{CH}_2=\text{CH}-\text{CN}$ . Its polymer backbone consists entirely of non-polar carbon-carbon ( $\text{C}-\text{C}$ ) sigma bonds. Because it lacks ester ( $-\text{COO}-$ ) or amide ( $-\text{CONH}-$ ) links in its main backbone, it cannot undergo hydrolysis by water, acids, bases, or microbial enzymes, persisting indefinitely in landfills.                                                                                                                                                                                                                                                             | [1]   |